SET 66 • HSG-GPE.A.2 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Derive the equation of a parabola from a focus and directrix

MAIN	Geometry main question	**HSG-GPE.A.2**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	Distance from a point to a horizontal line	**6.NS.C.8**
BACKWARD 3	Points equidistant from two objects	**HSG-CO.C.9**
BACKWARD 4	Expand and simplify squares	**HSA-APR.A.1**
FORWARD 1	Find the focus and directrix from an equation	**HSG-GPE.A.2**
FORWARD 2	Design with the reflective property of parabolas	**HSG-MG.A.3**

SET 66 • HSG-GPE.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Derive the equation of the parabola with focus (0, 2) and directrix y = −2.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q1/5

2403

QUESTION ID

S66-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x^2 = 8y$ ($y = x^2/8$)

GRADING NOTE REQUIRED WORK: Set the distance from (x, y) to the focus equal to its distance to the directrix, square, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $\sqrt{(x^2 + (y - 2)^2)} = |y + 2|$; square: $x^2 + y^2 - 4y + 4 = y^2 + 4y + 4$; simplify to $x^2 = 8y$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the parabola with focus (0, −3) and directrix y = 3.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q2/5

2404

QUESTION ID

S66-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 = -12y$

GRADING NOTE Form required: an equation; equivalent forms ($y = -x^2/12$) accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the parabola with focus (2, 5) and directrix y = 1.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q3/5

2405

QUESTION ID

S66-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(x - 2)^2 = 8(y - 3)$

GRADING NOTE Form required: an equation; equivalent forms accepted. Vertex (2, 3) midway; p = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the parabola with focus (3, 0) and directrix x = −3.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q4/5

2406

QUESTION ID

S66-M-Q4

TYPE

Short answer

CORRECT ANSWER

$y^2 = 12x$

GRADING NOTE Form required: an equation; equivalent forms accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q4/5

A parabola has focus (1, 4) and directrix y = 0. Its vertex is

A. (1, 4)

B. (0, 2)

C. (1, 0)

D. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q5/5

2407

QUESTION ID

S66-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 2)

GRADING NOTE Midway between focus and directrix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (3, 6) to (0, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q1/5

2409

QUESTION ID

S66-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the distance from (x, y) to (0, 2) as an expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q2/5

2410

QUESTION ID

S66-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$\sqrt{(x^2 + (y - 2)^2)}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The distance from (4, 1) to (1, 5) is

A. 7

B. 5

C. √7

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q3/5

2411

QUESTION ID

S66-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

GRADING NOTE Legs 3 and 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q3/5

True or false?

The distance formula comes from the Pythagorean Theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q4/5

2412

QUESTION ID

S66-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Legs are coordinate differences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{(9 + 16)} = 3 + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q5/5

2413

QUESTION ID

S66-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{25} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 2

Distance from a point to a horizontal line • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

How far is (5, 7) from the line y = 2?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q1/5

2415

QUESTION ID

S66-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How far is (1, −3) from the line y = 4?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q2/5

2416

QUESTION ID

S66-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q2/5

The distance from (x, y) to y = −2 is

A. |x + 2|

B. y

C. |y − 2|

D. |y + 2|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q3/5

2417

QUESTION ID

S66-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. |y + 2|

GRADING NOTE Vertical difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q3/5

True or false?

The distance from a point to a horizontal line is measured straight up or down.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q4/5

2418

QUESTION ID

S66-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q4/5

True or false?

The distance from (6, 1) to y = 4 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q5/5

2419

QUESTION ID

S66-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 3

Points equidistant from two objects • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

The set of points equidistant from two points is the perpendicular bisector of the segment joining them.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q1/5

2421

QUESTION ID

S66-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q1/5

A parabola is the set of points equidistant from what two objects?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q2/5

2422

QUESTION ID

S66-B3-Q2

TYPE

Short answer

CORRECT ANSWER

A point (the focus) and a line (the directrix)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q2/5

The vertex of a parabola is

A. at the focus

B. twice as far from the focus

C. halfway between the focus and the directrix

D. on the directrix

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q3/5

2423

QUESTION ID

S66-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. halfway between the focus and the directrix

GRADING NOTE Equidistant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Focus (0, 6), directrix y = 0. Find the vertex.

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q4/5

2424

QUESTION ID

S66-B3-Q4

TYPE

Short answer

CORRECT ANSWER

(0, 3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q4/5

True or false?

Every point of a parabola is closer to the focus than to the directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q5/5

2425

QUESTION ID

S66-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Equal distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 4

Expand and simplify squares • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(y - 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q1/5

2427

QUESTION ID
S66-B4-Q1

TYPE
Short answer

CORRECT ANSWER
$y^2 - 4y + 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(y + 2)^2 - (y - 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q2/5

2428

QUESTION ID
S66-B4-Q2

TYPE
Short answer

CORRECT ANSWER
8y

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$y^2 + 6y + 9 - (y^2 - 6y + 9) =$

A. 18

B. $2y^2$

C. 0

D. 12y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q3/5

2429

QUESTION ID
S66-B4-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 12y

GRADING NOTE The squares cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(y + 3)^2 = y^2 + 6y + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q4/5

2430

QUESTION ID
S66-B4-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(y - 2)^2 - y^2 = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q5/5

2431

QUESTION ID
S66-B4-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $-4y + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q5/5

SET 66 • HSG-GPE.A.2

FORWARD 1

Find the focus and directrix from an equation • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

Find the focus and directrix of $y = x^2/12$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q1/5

2433

QUESTION ID

S66-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Focus (0, 3); directrix y = −3

GRADING NOTE Form required: both. $x^2 = 12y = 4py$, $p = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q1/5

Find the vertex, focus, and directrix of $(x - 1)^2 = -8(y + 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q2/5

2434

QUESTION ID

S66-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Vertex (1, −2); focus (1, −4); directrix y = 0

GRADING NOTE Form required: all three. p = −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q2/5

True or false?

In $x^2 = 4py$, the focus is p units from the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q3/5

2435

QUESTION ID
S66-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q3/5

The parabola $y^2 = -20x$ opens

A. left

B. up

C. down

D. right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q4/5

2436

QUESTION ID

S66-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. left

GRADING NOTE Negative coefficient with y^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the focus of $y = 2x^2$.

Focus: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q5/5

2437

QUESTION ID

S66-F1-Q5

TYPE

Short answer

CORRECT ANSWER

(0, ⅛)

GRADING NOTE Form required: an ordered pair. $x^2 = \frac{1}{2}y$, $4p = \frac{1}{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q5/5

SET 66 • HSG-GPE.A.2

FORWARD 2

Design with the reflective property of parabolas • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A satellite dish is 4 ft across and 1 ft deep. How far from the vertex should the receiver (at the focus) be placed?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q1/5

2439

QUESTION ID

S66-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1 ft

GRADING NOTE Any equivalent form. $x^2 = 4py$ through (2, 1): $4 = 4p$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A headlight reflector has equation $x^2 = 12y$ (inches). Where should the bulb go?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q2/5

2440

QUESTION ID
S66-F2-Q2

TYPE
Short answer

CORRECT ANSWER
3 in. above the vertex, at (0, 3)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q2/5

True or false?

Rays parallel to the axis of a parabolic mirror reflect through the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q3/5

2441

QUESTION ID
S66-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They reflect through the focus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q3/5

A dish has focus 2 m above its vertex. Its equation (vertex at the origin) is

A. $x^2 = 4y$

B. $x^2 = 8y$

C. $y = 8x^2$

D. $x^2 = 2y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q4/5

2442

QUESTION ID

S66-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $x^2 = 8y$

GRADING NOTE $4p = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A dish has equation $x^2 = 16y$ (meters) and is 8 m across. How deep is it?

Depth: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q5/5

2443

QUESTION ID
S66-F2-Q5

TYPE
Short answer

CORRECT ANSWER
1 m

GRADING NOTE Any equivalent form. 16 = 16y at x = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q5/5

SET 67 • HSG-GPE.B.4 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Prove or disprove the type of a quadrilateral with coordinates

MAIN	Geometry main question	**HSG-GPE.B.4**
BACKWARD 1	Find the slope between two points	**8.F.B.4**
BACKWARD 2	Find the distance between two points	**8.G.B.8**
BACKWARD 3	Properties that define special quadrilaterals	**HSG-CO.C.11**
FORWARD 1	Coordinate proofs with variables	**HSG-GPE.B.4**
FORWARD 2	Use the classification to find area	**HSG-GPE.B.7**

SET 67 • HSG-GPE.B.4

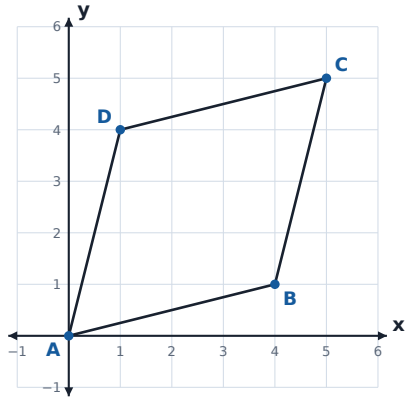
MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Prove that ABCD with A(0, 0), B(4, 1), C(5, 5), D(1, 4) is a rhombus.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q1/5

2446

QUESTION ID

S67-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AB = BC = CD = DA = √17, so ABCD is a rhombus.

GRADING NOTE REQUIRED WORK: Compute all four side lengths with the distance formula.

Any valid coordinate proof. Essential: all four distances computed correctly (√17 each) and the definition of a rhombus cited. Alternative: show it is a parallelogram (slopes) with perpendicular diagonals (slopes 1 and −1). Invalid: "it looks like a rhombus".

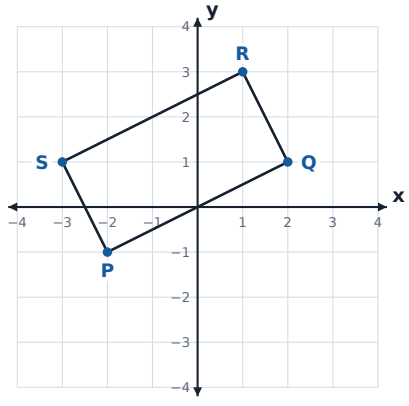
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that PQRS with P(−2, −1), Q(2, 1), R(1, 3), S(−3, 1) is a rectangle. Is it a square?

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q2/5

2447

QUESTION ID

S67-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Slopes PQ = RS = ½ and QR = SP = −2; adjacent products are −1, so all angles are right: a rectangle. Not a square: PQ = √20 but QR = √5.

GRADING NOTE REQUIRED WORK: Compute the slopes of all four sides; compare adjacent slopes.

Any valid coordinate proof. Essential: four slopes; opposite sides parallel; adjacent slopes multiply to −1 (right angles); side lengths √20 ≠ √5 for "not a square".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove: KLMN with K(0, 0), L(5, 0), M(6, 3), N(0, 3) is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q3/5

2448

QUESTION ID

S67-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Disproved: KL ∥ MN (slope 0), but LM has slope 3 while NK is vertical, so LM ∦ NK. KLMN is a trapezoid, not a parallelogram.

GRADING NOTE

REQUIRED WORK: Compare the slopes of both pairs of opposite sides.

Any valid coordinate proof. Essential: one pair of opposite sides shown NOT parallel (a single counterexample disproves). Alternative: KL = 5 but MN = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Classify the triangle with vertices A(1, 1), B(4, 5), C(8, 2) by its sides and angles. Justify.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q4/5

2449

QUESTION ID

S67-M-Q4

TYPE

Short answer

CORRECT ANSWER

Isosceles right triangle: AB = BC = 5 and AC = √50;
$5^2 + 5^2 = 50$, so the angle at B is right.

GRADING NOTE Form required: the classification with distances. Alternative: slopes AB = 4/3 and BC = −3/4 multiply to −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q4/5

A quadrilateral is shown to be a parallelogram. Which additional calculation proves it is a rectangle?

A. Two adjacent sides have slopes whose product is −1

B. Two opposite sides have equal slopes

C. The diagonals have equal slopes

D. Two adjacent sides have equal length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q5/5

2450

QUESTION ID

S67-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Two adjacent sides have slopes whose product is −1

GRADING NOTE A parallelogram with one right angle is a rectangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q5/5

SET 67 • HSG-GPE.B.4

BACKWARD 1

Find the slope between two points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (1, 2) and (5, 4).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q1/5

2452

QUESTION ID

S67-B1-Q1

TYPE

Short answer

CORRECT ANSWER

½

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (0, 3) and (2, −1).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q2/5

2453

QUESTION ID

S67-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The slope through (2, 5) and (6, 5) is

A. undefined

B. 0

C. 5

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q3/5

2454

QUESTION ID

S67-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0

GRADING NOTE Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q3/5

True or false?

A vertical segment has an undefined slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q4/5

2455

QUESTION ID

S67-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Run 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q4/5

True or false?

Slope is run divided by rise.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q5/5

2456

QUESTION ID

S67-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rise ÷ run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • HSG-GPE.B.4

BACKWARD 2

Find the distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (4, 1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q1/5

2458

QUESTION ID

S67-B2-Q1

TYPE

Short answer

CORRECT ANSWER

√17

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (1, 1) and (4, 5).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q2/5

2459

QUESTION ID

S67-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair of points is 10 apart?

A. (2, 3) and (8, 3)

B. (0, 0) and (6, 8)

C. (0, 0) and (10, 10)

D. (1, 1) and (5, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q3/5

2460

QUESTION ID

S67-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (0, 0) and (6, 8)

GRADING NOTE 36 + 64 = 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q3/5

True or false?

Two segments with the same length are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q4/5

2461

QUESTION ID

S67-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q4/5

True or false?

$\sqrt{10} + \sqrt{10} = \sqrt{20}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q5/5

2462

QUESTION ID

S67-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{10} + \sqrt{10} = 2\sqrt{10} = \sqrt{40}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSG-GPE.B.4

BACKWARD 3

Properties that define special quadrilaterals • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Which quadrilateral has four congruent sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q1/5

2464

QUESTION ID

S67-B3-Q1

TYPE

Short answer

CORRECT ANSWER

A rhombus (a square is a special rhombus)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram with one right angle is a ___.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q2/5

2465

QUESTION ID

S67-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Rectangle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q2/5

Which shows a quadrilateral is a parallelogram?

A. One pair of sides is parallel

B. Two adjacent sides are equal

C. One right angle

D. Both pairs of opposite sides are parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q3/5

2466

QUESTION ID

S67-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Both pairs of opposite sides are parallel

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q3/5

True or false?

Every square is a rhombus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q4/5

2467

QUESTION ID

S67-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Four congruent sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q4/5

True or false?

Every rhombus is a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q5/5

2468

QUESTION ID

S67-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its angles need not be right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q5/5

SET 67 • HSG-GPE.B.4

FORWARD 1

Coordinate proofs with variables • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has vertices O(0, 0), A(a, 0), B(a, b), C(0, b). Prove that its diagonals are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q1/5

2470

QUESTION ID

S67-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

OB = AC = √(a² + b²)

GRADING NOTE

REQUIRED WORK: Compute OB and AC with the distance formula.

Any valid coordinate proof. Essential: OB = √(a² + b²), AC = √((0 − a)² + (b − 0)²) = √(a² + b²); equal for all a, b. Invalid: using numbers for a and b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle with vertices O(0, 0), A(2a, 0), B(2b, 2c). M and N are the midpoints of OB and AB. Prove that MN ∥ OA and MN = ½OA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q2/5

2471

QUESTION ID

S67-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

M(b, c), N(a + b, c); MN is horizontal like OA, and MN = a = ½(2a).

GRADING NOTE REQUIRED WORK: Find M and N, then compare slope and length.

Any valid coordinate proof. Essential: correct midpoints; slope 0 for both; length a vs 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q2/5

True or false?

Placing a figure with a vertex at the origin and a side on the x-axis is allowed in a coordinate proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q3/5

2472

QUESTION ID

S67-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Any figure can be moved there by a rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q3/5

Why do coordinates like (2a, 0) and (2b, 2c) make the midpoint proof easier?

A. The midpoints have no fractions

B. They make the triangle isosceles

C. They make a right angle

D. They make the slopes 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q4/5

2473

QUESTION ID

S67-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The midpoints have no fractions

GRADING NOTE Halves of even coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q4/5

Why is a proof using A(6, 0) and B(2, 4) not enough to prove the midsegment theorem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q5/5

QUESTION ID

S67-F1-Q5

TYPE

Short answer

CORRECT ANSWER

It shows only one triangle; the theorem is about all triangles.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSG-GPE.B.4

FORWARD 2

Use the classification to find area • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ABCD with A(0, 0), B(4, 1), C(5, 5), D(1, 4) is a rhombus. Find its area using its diagonals.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q1/5

2476

QUESTION ID

S67-F2-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. AC = 5√2, BD = 3√2; ½ · 5√2 · 3√2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rectangle PQRS has P(−2, −1), Q(2, 1), R(1, 3), S(−3, 1). Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q2/5

2477

QUESTION ID

S67-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $\sqrt{20} \cdot \sqrt{5}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q2/5

True or false?

Once a quadrilateral is proved a rhombus, its area is the product of two adjacent side lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q3/5

2478

QUESTION ID

S67-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only for a square; use $\frac{1}{2}d_1d_2$ or base × height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q3/5

A square has vertices (0, 0), (3, 1), (2, 4), (−1, 3). Its area is

A. 16

B. √10

C. 10

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q4/5

2479

QUESTION ID

S67-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10

GRADING NOTE Side √10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why proving the shape first matters before using a special area formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q5/5

2480

QUESTION ID
S67-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The formula (such as ½d_1d_2 or length × width) is valid only for that type of figure.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q5/5

Prove or disprove that a point lies on a circle, and other coordinate proofs

MAIN	Geometry main question	**HSG-GPE.B.4**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	The equation of a circle	**HSG-GPE.A.1**
BACKWARD 3	Substitute values into an expression	**6.EE.A.2**
FORWARD 1	Coordinate proofs about circles with variables	**HSG-GPE.B.4**
FORWARD 2	Find the tangent line to a circle at a point	**HSG-GPE.B.5**

SET 68 • HSG-GPE.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A circle is centered at the origin and contains the point (0, 2). Prove or disprove that the point (1, √3) lies on the circle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q1/5

2483

QUESTION ID

S68-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Proved: the radius is 2 and the distance to (1, √3) is √(1 + 3) = 2.

GRADING NOTE REQUIRED WORK: Find the radius, then compare the distance from the origin to (1, √3).

Any valid coordinate proof. Essential: radius 2; distance $\sqrt{(1^2 + (\sqrt{3})^2)} = 2$; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove that (5, 2) lies on the circle with center (1, −1) and radius 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q2/5

2484

QUESTION ID

S68-M-Q2

TYPE

Short answer

CORRECT ANSWER

Proved:

$$(5 - 1)^2 + (2 + 1)^2 = 16 + 9 = 25 = 5^2.$$

GRADING NOTE Form required: the conclusion with the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove that $(-3, 6)$ lies on the circle $(x - 1)^2 + (y - 2)^2 = 30$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q3/5

2485

QUESTION ID

S68-M-Q3

TYPE

Short answer

CORRECT ANSWER

Disproved: 16 + 16 = 32 ≠ 30 (the point is outside).

GRADING NOTE Form required: the conclusion with the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has A(0, 0), B(8, 0), C(0, 6). Prove that the midpoint M of hypotenuse BC is equidistant from A, B, and C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q4/5

2486

QUESTION ID

S68-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

M(4, 3);

MA = MB = MC = 5

GRADING NOTE

REQUIRED WORK: Find M, then compute MA, MB, MC.

Any valid coordinate proof. Essential: midpoint (4, 3); three distances of 5. Consequence: A, B, C lie on a circle centered at M.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q4/5

The point (4, −3) is ___ the circle $x^2 + y^2 = 20$.

A. on

B. inside

C. outside

D. the center of

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q5/5

2487

QUESTION ID

S68-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. outside

GRADING NOTE 16 + 9 = 25 > 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q5/5

SET 68 • HSG-GPE.B.4

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (1, −1) to (5, 2)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q1/5

2489

QUESTION ID

S68-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (0, 0) to (1, √3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q2/5

2490

QUESTION ID

S68-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (0, 0) to (−3, 4) is

A. 5

B. 1

C. 7

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q3/5

2491

QUESTION ID

S68-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

GRADING NOTE 9 + 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q3/5

True or false?

$(-3)^2 = 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q4/5

2492

QUESTION ID

S68-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Square of a negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q4/5

True or false?

The distance from (2, 2) to (2, 7) is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q5/5

2493

QUESTION ID

S68-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSG-GPE.B.4

BACKWARD 2

The equation of a circle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the circle with center (1, −1) and radius 5.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q1/5

2495

QUESTION ID

S68-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 1)^2 + (y + 1)^2 = 25$

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the radius of $(x - 1)^2 + (y - 2)^2 = 30$?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q2/5

2496

QUESTION ID

S68-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$\sqrt{30}$

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Center of $(x + 4)^2 + y^2 = 9$?

A. (4, 0)

B. (−4, 0)

C. (−4, 9)

D. (0, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q3/5

2497

QUESTION ID

S68-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−4, 0)

GRADING NOTE Opposite sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q3/5

True or false?

A point satisfies a circle's equation exactly when it is on the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q4/5

2498

QUESTION ID

S68-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Meaning of the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q4/5

True or false?

The radius of $x^2 + y^2 = 20$ is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q5/5

2499

QUESTION ID

S68-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{20}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • HSG-GPE.B.4

BACKWARD 3

Substitute values into an expression • Grade 6 • Expressions & Equations • 6.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(x - 1)^2 + (y - 2)^2$ when $x = -3$ and $y = 6$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q1/5

2501

QUESTION ID

S68-B3-Q1

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $x^2 + y^2$ when $x = 4$, $y = -3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q2/5

2502

QUESTION ID

S68-B3-Q2

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x + 1)^2$ when $x = -4$ is

A. -9

B. 9

C. 25

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q3/5

2503

QUESTION ID

S68-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9

GRADING NOTE $(-3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(-5 - 1)^2 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q4/5

2504

QUESTION ID

S68-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-6)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(2 + 1)^2 = 2^2 + 1^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q5/5

2505

QUESTION ID

S68-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $9 \neq 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q5/5

SET 68 • HSG-GPE.B.4

FORWARD 1

Coordinate proofs about circles with variables • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A(−r, 0) and B(r, 0) are endpoints of a diameter of the circle $x^2 + y^2 = r^2$, and P(x, y) is any other point on the circle. Prove that ∠APB is a right angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q1/5

2507

QUESTION ID

S68-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

slope PA · slope PB = $y^2/(x^2 - r^2)$ = $y^2/(-y^2) = -1$, so PA ⊥ PB.

GRADING NOTE REQUIRED WORK: Compute the slopes of PA and PB and use $x^2 + y^2 = r^2$.

Any valid coordinate proof. Essential: slopes $y/(x + r)$ and $y/(x - r)$; product $y^2/(x^2 - r^2)$; substitute $x^2 - r^2 = -y^2$; product −1. (P is not A or B, so $y \neq 0$.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Show that if (a, b) is on $x^2 + y^2 = r^2$, then (b, a) and (−a, b) are too.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q2/5

2508

QUESTION ID

S68-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$b^2 + a^2 = r^2$ and $(-a)^2 + b^2 = r^2$, the same sum.

GRADING NOTE Any equivalent argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q2/5

True or false?

A coordinate proof with variables covers every case at once.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q3/5

2509

QUESTION ID
S68-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE General.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To prove four points lie on one circle, show they are

A. all the same distance from one point

B. the same distance from the origin only

C. vertices of a rectangle only

D. all on one line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q4/5

2510

QUESTION ID

S68-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. all the same distance from one point

GRADING NOTE Common center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove: A(1, 0), B(0, 1), C(−1, 0), D(0, −2) lie on one circle centered at the origin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q5/5

2511

QUESTION ID

S68-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Disproved: D is 2 from the origin, the others 1.

GRADING NOTE Form required: the conclusion with distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSG-GPE.B.4

FORWARD 2

Find the tangent line to a circle at a point • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Find the equation of the line tangent to $x^2 + y^2 = 25$ at (3, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q1/5

2513

QUESTION ID

S68-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$y = -\frac{3}{4}x + 25/4$

GRADING NOTE Form required: an equation; equivalent forms (3x + 4y = 25) accepted. Radius slope 4/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the tangent to $(x - 1)^2 + (y - 1)^2 = 25$ at $(4, 5)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q2/5

2514

QUESTION ID

S68-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$y - 5 = -\frac{3}{4}(x - 4)$

GRADING NOTE Form required: an equation; equivalent forms accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q2/5

True or false?

The tangent at P is parallel to the radius drawn to P.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q3/5

2515

QUESTION ID
S68-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is perpendicular to that radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q3/5

The radius to (0, 5) on $x^2 + y^2 = 25$ is vertical, so the tangent there is

A. $x = 0$

B. $x = 5$

C. $y = x + 5$

D. $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q4/5

2516

QUESTION ID

S68-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. y = 5

GRADING NOTE Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q4/5

Explain how the slope of the tangent at (a, b) on $x^2 + y^2 = r^2$ is found.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q5/5

2517

QUESTION ID

S68-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The radius has slope b/a, so the tangent has slope −a/b (negative reciprocal), for a, b ≠ 0.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q5/5

Prove the slope criteria for parallel and perpendicular lines

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Similar slope triangles give the same slope	**8.EE.B.6**
BACKWARD 2	Rotate a point 90° about the origin	**HSG-CO.A.5**
BACKWARD 3	Negative reciprocals	**7.NS.A.2**
FORWARD 1	Use the criteria to classify pairs of lines	**HSG-GPE.B.5**
FORWARD 2	Use the criteria inside a general coordinate proof	**HSG-GPE.B.4**

SET 69 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that two distinct nonvertical lines with the same slope m are parallel.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q1/5

2520

QUESTION ID

S69-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$mx + b_1 = mx + b_2$ gives $b_1 = b_2$, a contradiction; so the lines never meet and are parallel.

GRADING NOTE REQUIRED WORK: Write the lines as $y = mx + b_1$ and $y = mx + b_2$ with $b_1 \neq b_2$ and show there is no intersection.

Any valid proof (paragraph, two-column, or flow). Essential: equations with equal slopes and different intercepts; solving for an intersection leads to $b_1 = b_2$ (false); conclusion. Alternative: a translation maps one line to the other; or congruent slope triangles with corresponding angles equal.

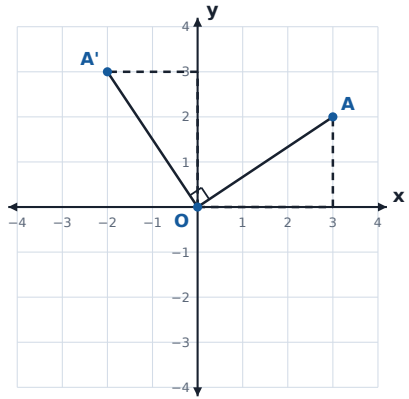
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Line OA passes through O(0, 0) and A(a, b) with a, b ≠ 0. Rotating A 90° counterclockwise about O gives A'(−b, a). Use this to prove that perpendicular lines have slopes whose product is −1. The diagram shows the case A(3, 2).

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q2/5

2521

QUESTION ID

S69-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

slope OA = b/a, slope OA' = a/(−b) = −a/b, product −1.

GRADING NOTE REQUIRED WORK: Find the slopes of OA and OA', then their product.

Any valid proof (paragraph, two-column, or flow). Essential: a 90° rotation maps OA onto a perpendicular line; the image of (a, b) is (−b, a); slopes b/a and −a/b; product −1. Any pair of perpendicular nonvertical lines can be translated to pass through O, keeping slopes. Invalid: checking only the one example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q2/5

Explain why a 90° rotation of the slope triangle with run 3 and rise 2 gives a slope triangle with run −2 and rise 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q3/5

2522

QUESTION ID

S69-M-Q3

TYPE

Short answer

CORRECT ANSWER

Rotating 90° counterclockwise sends (x, y) to (−y, x): the run becomes the old rise (negated) and the rise becomes the old run.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q3/5

Why do two nonvertical lines with equal slopes and different y-intercepts never meet?

A. Their slopes are negative reciprocals

B. Their intercepts are equal

C. Setting the equations equal leads to a false statement

D. They meet at the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q4/5

2523

QUESTION ID

S69-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Setting the equations equal leads to a false statement

GRADING NOTE $mx + b_1 = mx + b_2$ has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q4/5

True or false?

Lines with slopes 2 and −2 are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q5/5

2524

QUESTION ID

S69-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Product −4, not −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 1

Similar slope triangles give the same slope • Grade 8 • Expressions & Equations • 8.EE.B.6

True or false?

Any two slope triangles on the same line are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q1/5

2526

QUESTION ID

S69-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A line passes through (0, 0) and (2, 3). Find the rise for a run of 4.

Rise: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q2/5

2527

QUESTION ID

S69-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q2/5

Slope triangles with run 2, rise 3 and run 6, rise 9 give slopes

A. 6 and 18

B. 3/2 and 9/6 that differ

C. both 3/2

D. 2/3 and 6/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q3/5

2528

QUESTION ID

S69-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. both 3/2

GRADING NOTE 9/6 = 3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the slope of a line is the same between any two of its points.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q4/5

2529

QUESTION ID

S69-B1-Q4

TYPE

Short answer

CORRECT ANSWER

The slope triangles are similar, so rise/run is the same ratio.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q4/5

True or false?

The slope between two points on a line depends on which points are chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q5/5

2530

QUESTION ID

S69-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 2

Rotate a point 90° about the origin • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Rotate (3, 2) 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q1/5

2532

QUESTION ID

S69-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−2, 3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rotate (1, 4) 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q2/5

2533

QUESTION ID

S69-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−4, 1)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q2/5

The rule for a 90° counterclockwise rotation about the origin is

A. (x, y) → (−y, x)

B. (x, y) → (−x, −y)

C. (x, y) → (y, x)

D. (x, y) → (y, −x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q3/5

2534

QUESTION ID

S69-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (x, y) → (−y, x)

GRADING NOTE Rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q3/5

True or false?

A 90° rotation maps a line to a perpendicular line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q4/5

2535

QUESTION ID

S69-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Angle of rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q4/5

True or false?

A 90° rotation maps (2, 0) to (0, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q5/5

2536

QUESTION ID

S69-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Counterclockwise gives (0, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 3

Negative reciprocals • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Negative reciprocal of 2/3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q1/5

2538

QUESTION ID

S69-B3-Q1

TYPE

Short answer

CORRECT ANSWER

−3/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(b/a)(-a/b)$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q2/5

2539

QUESTION ID

S69-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Negative reciprocal of −4?

A. ¼

B. −¼

C. 4

D. −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q3/5

2540

QUESTION ID

S69-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ¼

GRADING NOTE −1 ÷ (−4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q3/5

True or false?

A number times its negative reciprocal is −1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q4/5

2541

QUESTION ID

S69-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The negative reciprocal of 1 is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q5/5

2542

QUESTION ID

S69-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q5/5

SET 69 • HSG-GPE.B.5

FORWARD 1

Use the criteria to classify pairs of lines • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Are y = 3x − 2 and 6x − 2y = 7 parallel, perpendicular, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q1/5

2544

QUESTION ID
S69-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Parallel (both slopes 3)

GRADING NOTE Form required: the classification with the slopes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q1/5

Are 2x + 5y = 1 and 5x − 2y = 4 parallel, perpendicular, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q2/5

2545

QUESTION ID

S69-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Perpendicular (slopes −2/5 and 5/2)

GRADING NOTE Form required: the classification with the slopes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q2/5

True or false?

A horizontal line and a vertical line are perpendicular even though the slope product rule cannot be used.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q3/5

2546

QUESTION ID

S69-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Vertical slope is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q3/5

$y = \frac{1}{2}x + 1$ and $y = 2x - 1$ are

A. neither

B. the same line

C. perpendicular

D. parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q4/5

2547

QUESTION ID

S69-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. neither

GRADING NOTE Product 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find k so that y = kx + 1 is perpendicular to y = −⅓x.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q5/5

2548

QUESTION ID

S69-F1-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q5/5

SET 69 • HSG-GPE.B.5

FORWARD 2

Use the criteria inside a general coordinate proof • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Rhombus OABC has O(0, 0), A(a, 0), B(a + b, c), C(b, c), with $b^2 + c^2 = a^2$. Prove that its diagonals are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q1/5

2550

QUESTION ID

S69-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

slope OB · slope AC = c/(a + b) · c/(b − a) = c²/(b² − a²) = c²/(−c²) = −1

GRADING NOTE REQUIRED WORK: Find the slopes of OB and AC and multiply.

Any valid coordinate proof. Essential: both slopes; product simplified with $b^2 + c^2 = a^2$; conclusion. (Assume $b \neq \pm a$, so neither diagonal is vertical.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A coordinate proof places rhombus OABC at O(0, 0), A(a, 0), B(a + b, c), C(b, c). Why must the proof assume $b^2 + c^2 = a^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q2/5

2551

QUESTION ID
S69-F2-Q2

TYPE
Short answer

CORRECT ANSWER
It says OC = a = OA, the equal sides that make OABC a rhombus.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slope criterion turns "perpendicular" into an algebra statement.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q3/5

2552

QUESTION ID
S69-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Product −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In a coordinate proof that lines are parallel, you show

A. equal slopes

B. equal intercepts

C. slope product −1

D. equal lengths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q4/5

2553

QUESTION ID

S69-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. equal slopes

GRADING NOTE Criterion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q4/5

Prove or disprove with slopes: A(0, 0), B(4, 2), C(3, 4) form a right angle at B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q5/5

2554

QUESTION ID

S69-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Proved: slope AB = ½ and slope BC = −2; the product is −1, so AB ⊥ BC.

GRADING NOTE Form required: the conclusion with slopes. Slope BC = (4 − 2)/(3 − 4) = −2; product −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q5/5

Write equations of parallel and perpendicular lines

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Read the slope of a line	**8.F.B.4**
BACKWARD 2	Rewrite an equation in slope-intercept form	**HSA-CED.A.4**
BACKWARD 3	Write a line from a point and a slope	**HSF-LE.A.2**
BACKWARD 4	Opposite reciprocals	**7.NS.A.2**
FORWARD 1	Write perpendicular bisectors and altitudes	**HSG-GPE.B.5**
FORWARD 2	Find the distance from a point to a line	**HSG-GPE.B.4**

SET 70 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line parallel to y = 3x − 5 that passes through (2, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q1/5

2557

QUESTION ID
S70-M-Q1

TYPE
Short answer

CORRECT ANSWER
y = 3x − 2

GRADING NOTE Form required: an equation; equivalent forms accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line perpendicular to $y = \frac{2}{3}x + 1$ that passes through $(4, -1)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q2/5

2558

QUESTION ID
S70-M-Q2

TYPE
Short answer

CORRECT ANSWER
y = −3/2 x + 5

GRADING NOTE Form required: an equation; equivalent forms accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line parallel to 2x + 5y = 10 that passes through (5, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q3/5

2559

QUESTION ID
S70-M-Q3

TYPE
Short answer

CORRECT ANSWER
y = −2/5 x + 5

GRADING NOTE Form required: an equation; equivalent forms (2x + 5y = 25) accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line perpendicular to x = 4 that passes through (1, −3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q4/5

2560

QUESTION ID
S70-M-Q4

TYPE
Short answer

CORRECT ANSWER
y = −3

GRADING NOTE Form required: an equation. x = 4 is vertical, so the line is horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which line is perpendicular to y = −4x + 1 and passes through the origin?

A. y = −4x

B. y = −¼x

C. y = ¼x

D. y = 4x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q5/5

2561

QUESTION ID

S70-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. y = ¼x

GRADING NOTE Negative reciprocal of −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 1

Read the slope of a line • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Slope of $y = -2x + 7$?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q1/5

2563

QUESTION ID

S70-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Slope of y = 5?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q2/5

2564

QUESTION ID

S70-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Slope of $y = \frac{2}{3}x - 4$?

A. 3/2

B. ⅔

C. −4

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q3/5

2565

QUESTION ID

S70-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. ⅔

GRADING NOTE Coefficient of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q3/5

True or false?

In y = mx + b, m is the slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q4/5

2566

QUESTION ID

S70-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Slope-intercept form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q4/5

True or false?

The slope of x = 3 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q5/5

2567

QUESTION ID

S70-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 2

Rewrite an equation in slope-intercept form • High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x + 5y = 10 for y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q1/5

2569

QUESTION ID

S70-B2-Q1

TYPE

Short answer

CORRECT ANSWER

y = −2/5 x + 2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3x − y = 4 for y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q2/5

2570

QUESTION ID

S70-B2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 3x − 4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

6x + 3y = 9 in slope-intercept form is

A. y = 2x + 3

B. y = −6x + 9

C. y = −2x + 9

D. y = −2x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q3/5

2571

QUESTION ID

S70-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = −2x + 3

GRADING NOTE Divide by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x - 2y = 6$ has slope ½.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q4/5

2572

QUESTION ID

S70-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $y = ½x - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

4x + 2y = 8 has slope 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q5/5

2573

QUESTION ID

S70-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 3

Write a line from a point and a slope • High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the line with slope 3 through (2, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q1/5

2575

QUESTION ID

S70-B3-Q1

TYPE

Short answer

CORRECT ANSWER

y = 3x − 2

GRADING NOTE Form required: an equation; equivalent forms accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B3-Q1**

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the line with slope −1 through (0, 6).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q2/5

2576

QUESTION ID

S70-B3-Q2

TYPE

Short answer

CORRECT ANSWER

y = −x + 6

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Slope 2 through (1, 5): b =

A. 5

B. 7

C. −3

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q3

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q3/5

2577

QUESTION ID

S70-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE 5 = 2 + b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B3-Q3**

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q3/5

True or false?

$y - 4 = 3(x - 2)$ is the line with slope 3 through (2, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S70-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q4/5

2578

QUESTION ID

S70-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Point-slope form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S70-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A line through (0, 6) has y-intercept 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q5/5

2579

QUESTION ID

S70-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 4

Opposite reciprocals • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Negative reciprocal of ⅔?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q1/5

2581

QUESTION ID

S70-B4-Q1

TYPE

Short answer

CORRECT ANSWER

−3/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Negative reciprocal of −5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q2/5

2582

QUESTION ID

S70-B4-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{5}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Negative reciprocal of −2/5?

A. 5/2

B. 2/5

C. −2/5

D. −5/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q3/5

2583

QUESTION ID

S70-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5/2

GRADING NOTE Flip and change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q3/5

True or false?

The negative reciprocal of ¼ is −4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q4/5

2584

QUESTION ID

S70-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Flip and change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q4/5

True or false?

The negative reciprocal of 3 is −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q5/5

2585

QUESTION ID

S70-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −⅓.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q5/5

SET 70 • HSG-GPE.B.5

FORWARD 1

Write perpendicular bisectors and altitudes • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Write the equation of the perpendicular bisector of the segment from (2, 1) to (6, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q1/5

2587

QUESTION ID

S70-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = −x + 7

GRADING NOTE Form required: an equation. Midpoint (4, 3), slope −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q1/5

Triangle ABC has A(0, 0), B(6, 0), C(2, 4). Write the equation of the altitude from A to BC.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q2/5

2588

QUESTION ID

S70-F1-Q2

TYPE

Short answer

CORRECT ANSWER

y = x

GRADING NOTE Form required: an equation. BC slope −1, so the altitude has slope 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q2/5

True or false?

A perpendicular bisector passes through the midpoint of the segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q3/5

2589

QUESTION ID

S70-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The perpendicular bisector of the segment from (0, 0) to (0, 8) is

A. x = 0

B. y = 4

C. x = 4

D. y = 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q4/5

2590

QUESTION ID

S70-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y = 4

GRADING NOTE Horizontal through (0, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the perpendicular bisector of the segment from (−3, 2) to (5, 2).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q5/5

2591

QUESTION ID

S70-F1-Q5

TYPE

Short answer

CORRECT ANSWER

x = 1

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q5/5

SET 70 • HSG-GPE.B.5

FORWARD 2

Find the distance from a point to a line • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from P(5, 5) to the line y = −x + 2.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q1/5

2593

QUESTION ID

S70-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

4√2 (about 5.66); the foot is (1, 1)

GRADING NOTE REQUIRED WORK: Write the perpendicular through P, find where it meets the line, then use the distance formula. Form required: exact value or rounding to 0.01. Perpendicular y = x meets y = −x + 2 at (1, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (0, 0) to the line y = 4.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q2/5

2594

QUESTION ID

S70-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q2/5

True or false?

The distance from a point to a line can be measured along any segment from the point to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q3/5

2595

QUESTION ID
S70-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is measured along the perpendicular, the shortest segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The foot of the perpendicular from (3, 0) to y = x is

A. (0, 3)

B. (0, 0)

C. (3, 3)

D. (1.5, 1.5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q4/5

2596

QUESTION ID

S70-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (1.5, 1.5)

GRADING NOTE y = −x + 3 meets y = x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (6, 1) to the line y = 2x − 1.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q5/5

2597

QUESTION ID

S70-F2-Q5

TYPE

Short answer

CORRECT ANSWER

2√5 (about 4.47)

GRADING NOTE Form required: exact value or rounding to 0.01. Foot (2, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q5/5

Solve geometric problems with the slope criteria

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Slope from two points	**8.F.B.4**
BACKWARD 2	Solve a linear equation for an unknown	**HSA-REI.B.3**
BACKWARD 3	Equations of parallel and perpendicular lines	**HSG-GPE.B.5**
BACKWARD 4	Opposite sides of a parallelogram are parallel	**HSG-CO.C.11**
FORWARD 1	Find the orthocenter of a triangle	**HSG-GPE.B.5**
FORWARD 2	Shortest path to a straight line in a design	**HSG-MG.A.3**

SET 71 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

The line through (1, 2) and (4, k) is parallel to $y = 2x + 7$. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q1/5

2600

QUESTION ID
S71-M-Q1

TYPE
Short answer

CORRECT ANSWER
8

GRADING NOTE Any equivalent form. $(k - 2)/3 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q1/5

ABCD is a parallelogram with A(0, 0), B(5, 1), C(7, 4). Find D.

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q2/5

2601

QUESTION ID
S71-M-Q2

TYPE
Short answer

CORRECT ANSWER
(2, 3)

GRADING NOTE Form required: an ordered pair. AD ∥ BC and DC ∥ AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q2/5

The segment from (2, 1) to (5, k) is perpendicular to the line y = 3x. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q3/5

2602

QUESTION ID
S71-M-Q3

TYPE
Short answer

CORRECT ANSWER
0

GRADING NOTE Any equivalent form. (k − 1)/3 = −⅓.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(8, 4), C(4, 6). Find the foot of the altitude from C to AB.

Foot: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q4/5

2603

QUESTION ID

S71-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

(5.6, 2.8)

GRADING NOTE REQUIRED WORK: Write line AB and the perpendicular through C, then solve the system.

Form required: an ordered pair. AB: y = ½x; altitude: y = −2x + 14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(1, 1), B(4, 2), C(3, 5). At which vertex is the right angle?

A. C

B. B

C. A

D. There is no right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q5/5

2604

QUESTION ID
S71-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. B

GRADING NOTE Slope AB = ⅓, slope BC = −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 1

Slope from two points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Slope through (1, 2) and (4, 8)?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q1/5

2606

QUESTION ID

S71-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Slope through (4, 2) and (3, 5)?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q2/5

2607

QUESTION ID

S71-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q2/5

Slope through (0, 0) and (8, 4)?

A. ½

B. 8

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q3/5

2608

QUESTION ID

S71-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ½

GRADING NOTE 4/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slope from A to B equals the slope from B to A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q4/5

2609

QUESTION ID

S71-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slope through (1, 1) and (1, 5) is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q5/5

2610

QUESTION ID

S71-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 2

Solve a linear equation for an unknown • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(k - 2)/3 = 2$.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q1/5

2612

QUESTION ID

S71-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-B2-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ½x = −2x + 14.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q2/5

2613

QUESTION ID

S71-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-B2-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(k - 1)/3 = -\frac{1}{3}$.

A. -2

B. 1

C. 2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q3/5

2614

QUESTION ID

S71-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE $k - 1 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q3/5

True or false?

Multiplying both sides by 3 is a valid first step in $(k - 2)/3 = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S71-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q4/5

2615

QUESTION ID

S71-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Inverse operation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S71-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q4/5

True or false?

The solution of 2x + 1 = 7 is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q5/5

2616

QUESTION ID

S71-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 3

Equations of parallel and perpendicular lines • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Line perpendicular to y = ½x through (4, 6)?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q1/5

2618

QUESTION ID

S71-B3-Q1

TYPE

Short answer

CORRECT ANSWER

y = −2x + 14

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Line parallel to y = −x through (0, 3)?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q2/5

2619

QUESTION ID

S71-B3-Q2

TYPE

Short answer

CORRECT ANSWER

y = −x + 3

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-B3-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Perpendicular slope to ⅓?

A. 3

B. −3

C. −⅓

D. ⅓

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q3/5

2620

QUESTION ID

S71-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −3

GRADING NOTE Negative reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q3/5

True or false?

An altitude is perpendicular to the side it meets.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q4/5

2621

QUESTION ID

S71-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q4/5

True or false?

An altitude passes through the midpoint of the opposite side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q5/5

2622

QUESTION ID

S71-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is a median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q5/5

BACKWARD 4

Opposite sides of a parallelogram are parallel • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

True or false?

In a parallelogram, both pairs of opposite sides are parallel and congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q1/5

2624

QUESTION ID

S71-B4-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram has A(0, 0), B(3, 0), C(4, 2). Find D (in order ABCD).

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q2/5

2625

QUESTION ID

S71-B4-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 2)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, side AB is parallel to

A. AC

B. BC

C. AD

D. DC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q3/5

2626

QUESTION ID

S71-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. DC

GRADING NOTE Opposite side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Going from B to C is "right 2, up 3". Going from A to D is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q4/5

2627

QUESTION ID

S71-B4-Q4

TYPE

Short answer

CORRECT ANSWER

Right 2, up 3

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A parallelogram's adjacent sides are always perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q5/5

2628

QUESTION ID

S71-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only rectangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q5/5

SET 71 • HSG-GPE.B.5

FORWARD 1

Find the orthocenter of a triangle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(6, 0), C(2, 4). Find the orthocenter (the intersection of the altitudes).

Orthocenter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q1/5

2630

QUESTION ID

S71-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

(2, 2)

GRADING NOTE REQUIRED WORK: Write two altitude equations and solve the system.

Form required: an ordered pair. Altitude from C: x = 2; from A: y = x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q1/5

Triangle with A(0, 0), B(4, 0), C(0, 3). Where is the orthocenter?

Orthocenter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q2/5

2631

QUESTION ID

S71-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 0)

GRADING NOTE Form required: an ordered pair. Right angle at A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q2/5

True or false?

The orthocenter of a right triangle is at the right-angle vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q3/5

2632

QUESTION ID

S71-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two legs are altitudes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To find the orthocenter, you need at least

A. three medians

B. the circumcenter

C. one altitude

D. two altitudes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q4/5

2633

QUESTION ID
S71-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. two altitudes

GRADING NOTE Two lines meet in a point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q4/5

2633

Check that the third altitude of the triangle A(0, 0), B(6, 0), C(2, 4) also passes through (2, 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q5/5

2634

QUESTION ID

S71-F1-Q5

TYPE

Short answer

CORRECT ANSWER

AC has slope 2, so the altitude from B has slope −½: y = −½(x − 6); at x = 2, y = 2. Yes.

GRADING NOTE Any equivalent check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSG-GPE.B.5

FORWARD 2

Shortest path to a straight line in a design • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A well is at W(2, 9). A straight canal runs along $y = 2x - 1$ (units: 100 m). Where should a pipe from the well meet the canal so the pipe is as short as possible, and how long is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q1/5

2636

QUESTION ID

S71-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

At (4.4, 7.8); about 2.68 units (268 m)

GRADING NOTE REQUIRED WORK: Use the perpendicular from W to the canal.

Form required: the point and the length (rounding to 0.01 unit or 1 m). Perpendicular $y = -\frac{1}{2}x + 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pipe will connect a well to a straight canal. Why is the shortest possible pipe perpendicular to the canal?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q2/5

2637

QUESTION ID
S71-F2-Q2

TYPE
Short answer

CORRECT ANSWER
The perpendicular segment is the shortest distance from a point to a line.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A pipe meeting the canal at a 60° angle is shorter than one meeting it at 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q3/5

2638

QUESTION ID

S71-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The perpendicular is shortest; the slanted pipe is a hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q3/5

2638

ANSWER KEY Reduced copy of the question page, with its answer

A path from (0, 5) to the road y = 0 is shortest at

A. (0, 0)

B. (5, 0)

C. (−5, 0)

D. (0, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q4/5

2639

QUESTION ID

S71-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (0, 0)

GRADING NOTE Straight down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A sensor at (3, 4) must connect to the line y = x. Where should the wire meet the line?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q5/5

2640

QUESTION ID
S71-F2-Q5

TYPE
Short answer

CORRECT ANSWER
(3.5, 3.5)

GRADING NOTE Form required: an ordered pair. y = −x + 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q5/5

2640

Partition a directed segment in a given ratio

MAIN	Geometry main question	**HSG-GPE.B.6**
BACKWARD 1	A ratio as a fraction of the whole	**6.RP.A.1**
BACKWARD 2	Horizontal and vertical change	**6.NS.C.8**
BACKWARD 3	Find a fraction of a number	**6.NS.A.1**
FORWARD 1	Work backward: find an endpoint or the ratio	**HSG-GPE.B.6**
FORWARD 2	The centroid as a partition point	**HSG-GPE.B.4**

SET 72 • HSG-GPE.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the point P on the directed segment from A(1, 2) to B(13, 8) that partitions it in the ratio 1 : 2.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q1/5

2643

QUESTION ID

S72-M-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 4)

GRADING NOTE Form required: an ordered pair. ⅓ of the way from A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the point P on the directed segment from A(−4, 6) to B(6, 1) that partitions it in the ratio 3 : 2.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q2/5

2644

QUESTION ID
S72-M-Q2

TYPE
Short answer

CORRECT ANSWER
(2, 3)

GRADING NOTE Form required: an ordered pair. 3/5 of the way from A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q2/5

Find the point P on the directed segment from A(2, −3) to B(−6, 9) that partitions it in the ratio 1 : 3.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q3/5

2645

QUESTION ID
S72-M-Q3

TYPE
Short answer

CORRECT ANSWER
(0, 0)

GRADING NOTE Form required: an ordered pair. ¼ of the way from A.

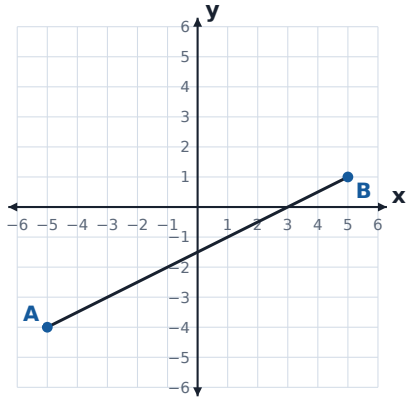
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q3/5

2645

ANSWER KEY Reduced copy of the question page, with its answer

Find the point P on the directed segment from A to B that partitions it in the ratio 2 : 3.



P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q4/5

2646

QUESTION ID

S72-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−1, −2)

GRADING NOTE Form required: an ordered pair. A(−5, −4), B(5, 1); 2/5 of (10, 5) is (4, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q4/5

2646

ANSWER KEY Reduced copy of the question page, with its answer

Point P partitions the directed segment from B(10, 5) to A(0, 0) in the ratio 1 : 4. P is

A. (8, 4)

B. (2, 1)

C. (5, 2.5)

D. (4, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q5/5

2647

QUESTION ID
S72-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. (8, 4)

GRADING NOTE Start at B: ⅕ of the way toward A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q5/5

SET 72 • HSG-GPE.B.6

BACKWARD 1

A ratio as a fraction of the whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A ratio of 2 : 3 splits a whole into how many equal parts?

Parts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q1/5

2649

QUESTION ID

S72-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a 1 : 2 partition, the first part is what fraction of the whole?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q2/5

2650

QUESTION ID

S72-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a 3 : 2 partition, the first part is

A. 3/2 of the whole

B. 2/3 of the whole

C. 3/5 of the whole

D. 2/5 of the whole

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q3/5

2651

QUESTION ID

S72-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/5 of the whole

GRADING NOTE 3 of 5 parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q3/5

2651

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A ratio of 1 : 3 means the point is ¼ of the way along.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q4/5

2652

QUESTION ID

S72-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1 of 4 parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A ratio of 1 : 2 means the point is halfway.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q5/5

2653

QUESTION ID

S72-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\frac{1}{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSG-GPE.B.6

BACKWARD 2

Horizontal and vertical change • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

From (1, 2) to (13, 8), what is the change in x?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q1/5

2655

QUESTION ID

S72-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

From (−4, 6) to (6, 1), what is the change in y?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q2/5

2656

QUESTION ID

S72-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−5

GRADING NOTE Any equivalent form (a decrease of 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q2/5

2656

ANSWER KEY Reduced copy of the question page, with its answer

From (2, −3) to (−6, 9), the change in x is

A. −8

B. −4

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q3/5

2657

QUESTION ID

S72-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −8

GRADING NOTE −6 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Change in x is the ending x minus the starting x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q4/5

2658

QUESTION ID

S72-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Direction matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q4/5

True or false?

The change in y from (0, 5) to (0, 1) is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q5/5

2659

QUESTION ID

S72-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSG-GPE.B.6

BACKWARD 3

Find a fraction of a number • Grade 6 • The Number System • 6.NS.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find ⅓ of 12.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q1/5

2661

QUESTION ID

S72-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find 3/5 of −5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q2/5

2662

QUESTION ID

S72-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

2/5 of 10 is

A. 5

B. 25

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q3/5

2663

GRADING NOTE 10 ÷ 5 × 2.

QUESTION ID

S72-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

¼ of −8 is −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q4/5

2664

QUESTION ID

S72-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

⅔ of 9 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q5/5

2665

QUESTION ID

S72-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q5/5

SET 72 • HSG-GPE.B.6

FORWARD 1

Work backward: find an endpoint or the ratio • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

P(3, 4) partitions the directed segment from A(1, 2) to B in the ratio 1 : 2. Find B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q1/5

2667

QUESTION ID

S72-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(7, 8)

GRADING NOTE Form required: an ordered pair. AP is ⅓ of AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q1/5

P(4, 3) is on the segment from A(0, 1) to B(10, 6). In what ratio does P partition AB?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q2/5

2668

QUESTION ID

S72-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2 : 3

GRADING NOTE Form required: a ratio. 4/10 = 2/5 of the way.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q2/5

True or false?

The midpoint partitions a segment in the ratio $1 : 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S72-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q3/5

2669

QUESTION ID
S72-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Halfway.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S72-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P partitions AB in 3 : 1. Then P is closer to

A. B

B. neither

C. A

D. the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q4/5

2670

QUESTION ID

S72-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. B

GRADING NOTE ¾ of the way from A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q4/5

M(2, 5) is the midpoint of AB with A(−1, 1). Find B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q5/5

2671

QUESTION ID

S72-F1-Q5

TYPE

Short answer

CORRECT ANSWER

(5, 9)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q5/5

SET 72 • HSG-GPE.B.6

FORWARD 2

The centroid as a partition point • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Triangle with A(0, 0), B(6, 0), C(3, 9). Find the point ⅔ of the way from A to the midpoint of BC.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q1/5

2673

QUESTION ID
S72-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(3, 3)

GRADING NOTE Form required: an ordered pair. Midpoint (4.5, 4.5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Show that the centroid (the point 2 : 1 along a median from the vertex) of the triangle with vertices (x_1, y_1), (x_2, y_2), (x_3, y_3) is the average of the vertices.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q2/5

2674

QUESTION ID

S72-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

$((x_1 + x_2 + x_3)/3,$
$(y_1 + y_2 + y_3)/3)$

GRADING NOTE REQUIRED WORK: Partition the median from (x_1, y_1) to the midpoint of the opposite side in the ratio 2 : 1.

Any valid coordinate proof. Essential: midpoint $((x_2 + x_3)/2, (y_2 + y_3)/2)$; $x_1 + \frac{2}{3}((x_2 + x_3)/2 - x_1) = (x_1 + x_2 + x_3)/3$; the same for y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q2/5

True or false?

The centroid of the triangle (0, 0), (6, 0), (3, 9) is (3, 4.5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q3/5

2675

QUESTION ID
S72-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is (3, 3), the average of the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The centroid of (1, 2), (4, 8), (7, −1) is

A. (4, 3)

B. (4, 9)

C. (3, 3)

D. (12, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q4/5

2676

QUESTION ID

S72-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (4, 3)

GRADING NOTE (12/3, 9/3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the centroid of the triangle with vertices (−2, 5), (4, 1), (7, 0).

Centroid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q5/5

2677

QUESTION ID
S72-F2-Q5

TYPE
Short answer

CORRECT ANSWER
(3, 2)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q5/5

SET 73 • HSG-GPE.B.7 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Compute perimeters of polygons with coordinates

MAIN	Geometry main question	**HSG-GPE.B.7**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	Lengths of horizontal and vertical sides	**6.G.A.3**
BACKWARD 3	Simplify square roots	**HSN-RN.A.2**
FORWARD 1	Find a missing coordinate from a perimeter	**HSG-GPE.B.7**
FORWARD 2	Fencing and cost from a coordinate plan	**HSG-MG.A.3**

SET 73 • HSG-GPE.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of the triangle with vertices (0, 0), (3, 4), and (3, 0).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q1/5

2680

QUESTION ID
S73-M-Q1

TYPE
Short answer

CORRECT ANSWER
12 units

GRADING NOTE Any equivalent form. 5 + 4 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of the quadrilateral with vertices A(1, 1), B(4, 5), C(10, −3), D(7, −7).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q2/5

2681

QUESTION ID
S73-M-Q2

TYPE
Short answer

CORRECT ANSWER
30 units

GRADING NOTE Any equivalent form. 5 + 10 + 5 + 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q2/5

Find the perimeter of the triangle with vertices (0, 0), (4, 0), (2, 4). Give the exact value and a decimal to the nearest tenth.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q3/5

2682

QUESTION ID

S73-M-Q3

TYPE

Short answer

CORRECT ANSWER

$4 + 4\sqrt{5} \approx 12.9$ units

GRADING NOTE Form required: both. Two sides $\sqrt{20} = 2\sqrt{5}$.

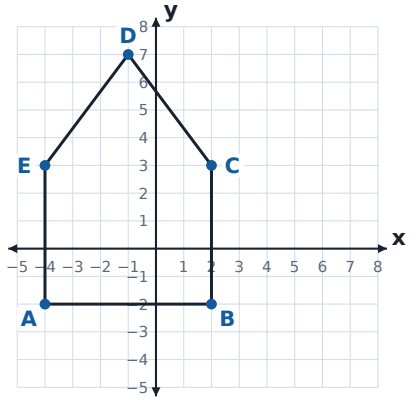
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of pentagon ABCDE.

Perimeter: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q4/5

2683

QUESTION ID
S73-M-Q4

TYPE
Short answer

CORRECT ANSWER
26 units

GRADING NOTE Any equivalent form. 6 + 5 + 5 + 5 + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A square has vertices (0, 0), (3, 1), (2, 4), (−1, 3). Its perimeter is

A. 4√10

B. 10

C. 40

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q5/5

2684

QUESTION ID
S73-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. 4√10

GRADING NOTE Side √10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (2, 3) to (−1, 7)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q1/5

2686

QUESTION ID

S73-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (4, 5) to (10, −3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q2/5

2687

QUESTION ID

S73-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q2/5

Distance from (0, 0) to (2, 4)?

A. 20

B. 2√5

C. 6

D. √6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q3/5

2688

QUESTION ID

S73-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2√5

GRADING NOTE √20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q3/5

True or false?

The distance from (0, 0) to (3, 1) is √10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q4/5

2689

QUESTION ID

S73-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 9 + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q4/5

True or false?

Distances can be negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q5/5

2690

QUESTION ID

S73-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Never.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 2

Lengths of horizontal and vertical sides • Grade 6 • Geometry • 6.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Length of the side from (−4, −2) to (2, −2)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q1/5

2692

QUESTION ID

S73-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Length of the side from (2, −2) to (2, 3)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q2/5

2693

QUESTION ID

S73-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q2/5

Length from $(-3, 4)$ to $(-3, -1)$?

A. 7

B. 3

C. 5

D. -5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q3/5

2694

QUESTION ID

S73-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5

GRADING NOTE $|4 - (-1)|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q3/5

True or false?

A horizontal side's length is the difference of the x-coordinates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q4/5

2695

QUESTION ID

S73-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Absolute difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q4/5

True or false?

The side from (1, 1) to (1, 9) has length 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q5/5

2696

QUESTION ID

S73-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 3

Simplify square roots • High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Simplify √20.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q1/5

2698

QUESTION ID
S73-B3-Q1

TYPE
Short answer

CORRECT ANSWER
2√5

GRADING NOTE Form required: simplest radical form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-B3-Q1**

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify √50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q2/5

2699

QUESTION ID
S73-B3-Q2

TYPE
Short answer

CORRECT ANSWER
5√2

GRADING NOTE Form required: simplest radical form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-B3-Q2**

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$2\sqrt{5} + 2\sqrt{5} =$

A. $2\sqrt{10}$

B. $8\sqrt{5}$

C. $4\sqrt{10}$

D. $4\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q3/5

2700

QUESTION ID

S73-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $4\sqrt{5}$

GRADING NOTE Like terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q3/5

True or false?

$\sqrt{10} \approx 3.16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q4/5

2701

QUESTION ID

S73-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Calculator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{9} + \sqrt{16} = \sqrt{25}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q5/5

2702

QUESTION ID

S73-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3 + 4 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q5/5

SET 73 • HSG-GPE.B.7

FORWARD 1

Find a missing coordinate from a perimeter • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has vertices (0, 0), (k, 0), (k, 3), (0, 3) with k > 0 and perimeter 22. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q1/5

2704

QUESTION ID

S73-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. 2k + 6 = 22.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle with vertices (0, 0), (6, 0), (3, h), h > 0, has perimeter 16. Find h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q2/5

2705

QUESTION ID
S73-F1-Q2

TYPE
Short answer

CORRECT ANSWER
4

GRADING NOTE Any equivalent form. Each slanted side is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q2/5

True or false?

Doubling every coordinate of a polygon multiplies its perimeter by 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q3/5

2706

QUESTION ID

S73-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The perimeter doubles; the area is multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A square with one side from (0, 0) to (4, 3) has perimeter

A. 28

B. 25

C. 14

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q4/5

2707

QUESTION ID

S73-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 20

GRADING NOTE Side 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q4/5

Is the triangle (0, 0), (6, 0), (3, 4) isosceles? Justify with lengths.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q5/5

2708

QUESTION ID

S73-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Yes: two sides are 5.

GRADING NOTE Form required: the conclusion with lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q5/5

SET 73 • HSG-GPE.B.7

FORWARD 2

Fencing and cost from a coordinate plan • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A garden plan has corners (0, 0), (8, 0), (8, 6), (0, 6), in meters. Fencing costs $12 per meter. Find the cost to fence it.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q1/5

2710

QUESTION ID
S73-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$336

GRADING NOTE Any equivalent form. 28 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangular lot has corners (0, 0), (30, 0), (30, 40) in feet. Edging costs $2.50 per foot. Find the total cost.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q2/5

2711

QUESTION ID

S73-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$300

GRADING NOTE Any equivalent form. 120 ft.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

On a map with 1 unit = 5 m, an area of 12 square units is 60 m².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q3/5

2712

QUESTION ID

S73-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Areas scale by 5²: 12 × 25 = 300 m².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A path has corners (0, 0), (3, 4), (9, 12) (units: 10 m). Its length is

A. 150 m

B. 210 m

C. 15 m

D. 21 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q4/5

2713

QUESTION ID
S73-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 150 m

GRADING NOTE 5 + 10 = 15 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A run around a lot with corners (0, 0), (5, 12), (5, 0) in units of 100 m: how many meters?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q5/5

2714

QUESTION ID
S73-F2-Q5

TYPE
Short answer

CORRECT ANSWER
3000 m

GRADING NOTE Any equivalent form. 13 + 12 + 5 = 30 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q5/5

Compute areas of triangles and rectangles with coordinates

MAIN	Geometry main question	**HSG-GPE.B.7**
BACKWARD 1	Area of a triangle from base and height	**6.G.A.1**
BACKWARD 2	Lengths on the coordinate plane	**6.G.A.3**
BACKWARD 3	Distance between two points	**8.G.B.8**
FORWARD 1	Find a missing vertex from an area	**HSG-GPE.B.7**
FORWARD 2	Use area from coordinates with density	**HSG-MG.A.2**

SET 74 • HSG-GPE.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of the triangle with vertices (0, 0), (8, 0), and (3, 5).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q1/5

2717

QUESTION ID
S74-M-Q1

TYPE
Short answer

CORRECT ANSWER
20 square units

GRADING NOTE Any equivalent form. Base 8, height 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rectangle ABCD has A(1, 1), B(5, 3), C(4, 5), D(0, 3). Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q2/5

2718

QUESTION ID
S74-M-Q2

TYPE
Short answer

CORRECT ANSWER
10 square units

GRADING NOTE Any equivalent form. AB = √20, BC = √5.

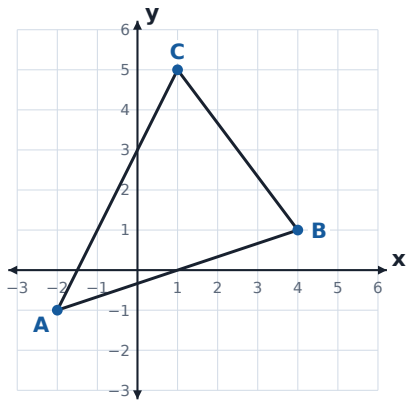
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of triangle ABC.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q3/5

2719

QUESTION ID
S74-M-Q3

TYPE
Short answer

CORRECT ANSWER
15 square units

GRADING NOTE Any equivalent form. Bounding rectangle 36 minus 6, 6, and 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of the triangle with vertices A(0, 0), B(6, 2), C(2, 6).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q4/5

2720

QUESTION ID

S74-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

16 square units

GRADING NOTE REQUIRED WORK: Enclose the triangle in a rectangle and subtract the three right triangles.

Any equivalent form. 36 − 6 − 8 − 6. Alternative methods (shoelace) accepted with work.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q4/5

A triangle has vertices (0, 0), (10, 0), and (x, 4). What is its area?

A. 20, for any x

B. It depends on x

C. 40

D. 20 only if x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q5/5

2721

QUESTION ID

S74-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 20, for any x

GRADING NOTE Base 10, height 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q5/5

SET 74 • HSG-GPE.B.7

BACKWARD 1

Area of a triangle from base and height • Grade 6 • Geometry • 6.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Area of a triangle with base 8 and height 5?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q1/5

2723

QUESTION ID

S74-B1-Q1

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Area of a right triangle with legs 6 and 2?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q2/5

2724

QUESTION ID

S74-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A =

A. ½bh

B. b + h

C. 2bh

D. bh

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q3/5

2725

QUESTION ID

S74-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ½bh

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q3/5

True or false?

The height of a triangle is perpendicular to the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q4/5

2726

QUESTION ID

S74-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The height must be one of the sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q5/5

2727

QUESTION ID

S74-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only in right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q5/5

SET 74 • HSG-GPE.B.7

BACKWARD 2

Lengths on the coordinate plane • Grade 6 • Geometry • 6.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Length from (−2, −1) to (4, −1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q1/5

2729

QUESTION ID

S74-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Height of the point (3, 5) above the x-axis?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q2/5

2730

QUESTION ID

S74-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q2/5

Width of the rectangle enclosing x from −2 to 4?

A. 2

B. 8

C. 4

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q3/5

2731

QUESTION ID

S74-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6

GRADING NOTE 4 − (−2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q3/5

True or false?

A point with y-coordinate −3 is 3 units from the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q4/5

2732

QUESTION ID

S74-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Absolute value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q4/5

True or false?

The length from (0, 2) to (0, 9) is 11.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q5/5

2733

QUESTION ID

S74-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • HSG-GPE.B.7

BACKWARD 3

Distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (1, 1) to (5, 3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q1/5

2735

QUESTION ID

S74-B3-Q1

TYPE

Short answer

CORRECT ANSWER

√20

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (5, 3) to (4, 5)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q2/5

2736

QUESTION ID

S74-B3-Q2

TYPE

Short answer

CORRECT ANSWER

√5

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sqrt{20} \cdot \sqrt{5} =$

A. 10

B. 25

C. $\sqrt{25}$

D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q3/5

2737

QUESTION ID

S74-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE $\sqrt{100}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q3/5

True or false?

Slanted rectangle sides need the distance formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q4/5

2738

QUESTION ID

S74-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Not grid-aligned.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{20} \cdot \sqrt{5} = 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q5/5

2739

QUESTION ID

S74-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q5/5

SET 74 • HSG-GPE.B.7

FORWARD 1

Find a missing vertex from an area • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Triangle with vertices (0, 0), (6, 0), (2, k) has area 18.
Find all possible k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q1/5

2741

QUESTION ID
S74-F1-Q1

TYPE
Short answer

CORRECT ANSWER
6 or −6

GRADING NOTE Form required: both values. ½ · 6 · |k| = 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has vertices (0, 0), (a, 0), (a, 5), (0, 5), $a > 0$, and area 35. Find a.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q2/5

2742

QUESTION ID
S74-F1-Q2

TYPE
Short answer

CORRECT ANSWER
7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q2/5

True or false?

Moving the top vertex of a triangle parallel to its base does not change its area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q3/5

2743

QUESTION ID

S74-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q3/5

Triangle (0, 0), (4, 0), (x, 6). For which x is the area 12?

A. Any x

B. No x

C. x = 4 only

D. x = 2 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q4/5

2744

QUESTION ID
S74-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. Any x

GRADING NOTE ½ · 4 · 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle (1, 1), (7, 1), (4, k) has area 21 and k > 1. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q5/5

2745

QUESTION ID

S74-F1-Q5

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. Base 6, height 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q5/5

SET 74 • HSG-GPE.B.7

FORWARD 2

Use area from coordinates with density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A park on a map has corners (0, 0), (4, 0), (4, 3), (0, 3), with 1 unit = 1 km. 1,800 people live there. Find the population density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q1/5

2747

QUESTION ID

S74-F2-Q1

TYPE

Short answer

CORRECT ANSWER

150 people per km²

GRADING NOTE Any equivalent form. Area 12 km².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangular field has vertices (0, 0), (200, 0), (0, 150) in meters. Seeds are planted at 4 per m². How many seeds?

Seeds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q2/5

2748

QUESTION ID
S74-F2-Q2

TYPE
Short answer

CORRECT ANSWER
60,000

GRADING NOTE Any equivalent form. Area 15,000 m².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q2/5

True or false?

Density based on area is an area divided by a quantity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q3/5

2749

QUESTION ID

S74-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is a quantity divided by an area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q3/5

A lot with corners (0, 0), (6, 0), (6, 5), (0, 5) (meters) holds 60 plants. Density?

A. 2 plants per m²

B. 30 plants per m²

C. 11 plants per m²

D. 0.5 plants per m²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q4/5

2750

QUESTION ID

S74-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2 plants per m²

GRADING NOTE 60/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle with corners (0, 0), (3, 0), (3, 2), (0, 2) (miles) has 4,200 residents. Find the density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q5/5

2751

QUESTION ID
S74-F2-Q5

TYPE
Short answer

CORRECT ANSWER
700 people per square mile

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q5/5

SET 75 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the circumference and area of a circle

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Circumference and area formulas	**7.G.B.4**
BACKWARD 2	Area of a parallelogram	**6.G.A.1**
BACKWARD 3	All circles are similar	**HSG-C.A.1**
FORWARD 1	Area of a circle as a limit of regular polygons	**HSG-GMD.A.1**
FORWARD 2	From area to volume: stacking disks	**HSG-GMD.A.1**

SET 75 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A circle of radius r is cut into many equal sectors, which are rearranged as shown. Explain why this shows that the area of the circle is πr^2.

Work space



Explanation: _____

Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q1/5

2754

QUESTION ID

S75-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Area = (half the circumference) × r = $\pi r \cdot r = \pi r^2$.

GRADING NOTE REQUIRED WORK: Describe the base and height of the rearranged shape and what happens as the number of sectors grows.

Any valid explanation. Essential: the shape approaches a parallelogram (rectangle); its height approaches r; its base is half the circumference, πr; area = $\pi r \cdot r = \pi r^2$. Invalid: "the formula says so" or measuring one drawing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the ratio of circumference to diameter is the same number (π) for every circle, so that $C = \pi d = 2\pi r$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q2/5

2755

QUESTION ID

S75-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Every circle is a dilation of every other, so C and d scale by the same factor and C/d is constant; that constant is called π.

GRADING NOTE

REQUIRED WORK: Use the fact that all circles are similar.

Any valid explanation. Essential: circles are similar (dilation); lengths scale by the same factor k; $C_2/d_2 = kC_1/kd_1 = C_1/d_1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q2/5

A circle of radius r is cut into many equal sectors, which are rearranged, alternating point up and point down, into a near-parallelogram. Its base gets closer and closer to

A. πr^2

B. $2\pi r$ (the whole circumference)

C. r

D. πr (half the circumference)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q3/5

2756

QUESTION ID

S75-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. πr (half the circumference)

GRADING NOTE Half the arcs are on top, half on the bottom.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A regular hexagon is inscribed in a circle of radius 5. Find the hexagon's perimeter and compare it with the circle's circumference. What happens as the number of sides increases?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q4/5

2757

QUESTION ID

S75-M-Q4

TYPE

Short answer

CORRECT ANSWER

Perimeter 30 < 10π ≈ 31.4; with more sides the perimeter gets closer to the circumference.

GRADING NOTE Form required: both numbers and the trend. The hexagon is six equilateral triangles of side 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q4/5

True or false?

As the number of sectors increases, the rearranged shape gets closer to a rectangle with height r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q5/5

2758

QUESTION ID

S75-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The scalloped edges flatten.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 1

Circumference and area formulas • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact circumference of a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q1/5

2760

QUESTION ID

S75-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact area of a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q2/5

2761

QUESTION ID

S75-B1-Q2

TYPE

Short answer

CORRECT ANSWER

16π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q2/5

Half the circumference of a circle of radius r is

A. πr^2

B. πr

C. $2\pi r$

D. r/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q3/5

2762

QUESTION ID

S75-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. πr

GRADING NOTE ½ · 2πr.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q3/5

True or false?

$C = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q4/5

2763

QUESTION ID

S75-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q4/5

True or false?

$A = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q5/5

2764

QUESTION ID

S75-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $A = \pi r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 2

Area of a parallelogram • Grade 6 • Geometry • 6.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Area of a parallelogram with base 9 and height 4?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q1/5

2766

QUESTION ID

S75-B2-Q1

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A parallelogram has the same area as a rectangle with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q2/5

2767

QUESTION ID

S75-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cut and slide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q2/5

A parallelogram's area is

A. base × slanted side

B. base × height

C. base + height

D. ½ base × height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q3/5

2768

QUESTION ID

S75-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. base × height

GRADING NOTE Perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram has base πr and height r. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q4/5

2769

QUESTION ID

S75-B2-Q4

TYPE

Short answer

CORRECT ANSWER

πr^2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slanted side is the height of a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q5/5

2770

QUESTION ID

S75-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 3

All circles are similar • High School — Geometry • Geometry: Circles • HSG-C.A.1

True or false?

Any circle can be mapped to any other circle by a translation and a dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q1/5

2772

QUESTION ID

S75-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle of radius 2 is dilated to radius 6. By what factor is the circumference multiplied?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q2/5

2773

QUESTION ID

S75-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Circles of radii 3 and 12 have circumferences in the ratio

A. 1 : 16

B. 3 : 4

C. 1 : 4

D. 1 : 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q3/5

2774

QUESTION ID

S75-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1 : 4

GRADING NOTE Scale factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q3/5

Why does C/d not depend on the size of the circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q4/5

2775

QUESTION ID

S75-B3-Q4

TYPE

Short answer

CORRECT ANSWER

Both C and d are multiplied by the same scale factor.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q4/5

True or false?

A larger circle has a larger value of π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q5/5

2776

QUESTION ID

S75-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE π is constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q5/5

SET 75 • HSG-GMD.A.1

FORWARD 1

Area of a circle as a limit of regular polygons • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A regular n-gon is inscribed in a circle of radius r. Its area is ½ · perimeter · apothem. Explain why this approaches πr² as n increases.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q1/5

2778

QUESTION ID

S75-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

The perimeter approaches 2πr and the apothem approaches r, so the area approaches ½ · 2πr · r = πr².

GRADING NOTE REQUIRED WORK: Describe what the perimeter and apothem approach.

Any valid explanation. Essential: perimeter → circumference; apothem → radius; substitution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of a regular hexagon with side 2 and apothem √3. Compare with the area of its circumscribed circle (radius 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q2/5

2779

QUESTION ID
S75-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$6\sqrt{3} \approx 10.4 < 4\pi \approx 12.6$

GRADING NOTE Form required: both numbers. ½ · 12 · √3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q2/5

True or false?

An inscribed polygon always has less area than its circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q3/5

2780

QUESTION ID

S75-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It lies inside the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

As n grows, the apothem of an inscribed regular n-gon approaches

A. πr

B. r

C. 0

D. 2r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q4/5

2781

QUESTION ID
S75-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. r

GRADING NOTE The sides approach the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q4/5

Why do mathematicians call this an "informal" argument?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q5/5

2782

QUESTION ID
S75-F1-Q5

TYPE
Short answer

CORRECT ANSWER
It relies on what the shapes approach (a limit) without a formal proof of the limit.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q5/5

SET 75 • HSG-GMD.A.1

FORWARD 2

From area to volume: stacking disks • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A stack of 40 identical coins, each a disk of radius 1 cm and thickness 0.2 cm, forms a cylinder. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q1/5

2784

QUESTION ID

S75-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8π cm^3

GRADING NOTE Form required: exact value. $\pi \cdot 1^2 \cdot 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A cylinder can be thought of as a stack of congruent disks.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q2/5

2785

QUESTION ID

S75-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Layers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The volume of a stack of disks of area πr^2 and total height h is

A. $\frac{1}{3}\pi r^2 h$

B. $2\pi rh$

C. $\pi r^2 h$

D. πr^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q3/5

2786

QUESTION ID
S75-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. $\pi r^2 h$

GRADING NOTE Base area × height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how the area of a circle is used to find the volume of a cylinder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q4/5

2787

QUESTION ID

S75-F2-Q4

TYPE

Short answer

CORRECT ANSWER

The volume is the base area πr^2 times the height h.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Each disk in a stack has area 12 cm² and the stack is 5 cm high. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q5/5

2788

QUESTION ID
S75-F2-Q5

TYPE
Short answer

CORRECT ANSWER
60 cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q5/5

SET 76 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal argument for the volume of a cylinder

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Volume of a rectangular prism as layers	**5.MD.C.5**
BACKWARD 2	Area of a circle	**7.G.B.4**
BACKWARD 3	Cross-sections parallel to the base	**7.G.A.3**
FORWARD 1	Cavalieri's principle with different shapes	**HSG-GMD.A.1**
FORWARD 2	Use the cylinder volume formula to solve problems	**HSG-GMD.A.3**

SET 76 • HSG-GMD.A.1

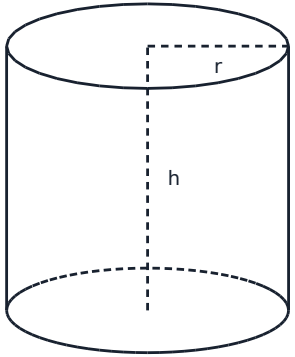
MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the volume of the cylinder shown is $V = \pi r^2 h$.

Work space



Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q1/5

2791

QUESTION ID

S76-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Each layer is a disk of area πr^2; stacking layers to height h gives volume (base area) × (height) = $\pi r^2 h$.

GRADING NOTE

REQUIRED WORK: Think of the cylinder as a stack of thin congruent disks.

Any valid explanation. Essential: congruent circular cross-sections of area πr^2; volume = base area × height, as for a prism (layers of unit thickness); result $\pi r^2 h$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A right cylinder and an oblique cylinder have the same radius r and the same height h. Explain why they have the same volume.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q2/5

2792

QUESTION ID

S76-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

At every height both cross-sections are disks of area πr², so by Cavalieri's principle their volumes are equal: πr²h.

GRADING NOTE REQUIRED WORK: Use Cavalieri's principle.

Any valid explanation. Essential: equal heights; equal cross-sectional areas at every level; Cavalieri's principle cited. Invalid: "they look the same size".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q2/5

A stack of 50 coins, each of radius 1 cm and thickness 0.2 cm, has volume

A. 10π cm³

B. 0.2π cm³

C. 50π cm³

D. 100π cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q3/5

2793

QUESTION ID

S76-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10π cm³

GRADING NOTE $\pi \cdot 1 \cdot 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why V = Bh works for both prisms and cylinders.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q4/5

2794

QUESTION ID

S76-M-Q4

TYPE

Short answer

CORRECT ANSWER

Both have congruent cross-sections of area B at every height, so the volume is B times the height.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q4/5

True or false?

Pushing a straight stack of coins into a slanted stack changes its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q5/5

2795

QUESTION ID
S76-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Same coins, same volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 1

Volume of a rectangular prism as layers • Grade 5 • Measurement & Data • 5.MD.C.5

ANSWER KEY Reduced copy of the question page, with its answer

A prism has a 4 × 3 base and height 5. How many unit cubes are in one layer, and what is the volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q1/5

2797

QUESTION ID

S76-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12 per layer; 60

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q1/5

True or false?

V = Bh, where B is the area of the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q2/5

2798

QUESTION ID

S76-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Layers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A prism has base area 20 and height 7. Volume?

A. 27

B. 140

C. 200

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q3/5

2799

QUESTION ID

S76-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 140

GRADING NOTE B × h.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q3/5

2799

ANSWER KEY Reduced copy of the question page, with its answer

A prism has 6 layers of 10 cubes each. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q4/5

2800

QUESTION ID

S76-B1-Q4

TYPE

Short answer

CORRECT ANSWER

60 cubic units

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q4/5

True or false?

Volume is measured in square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q5/5

2801

QUESTION ID

S76-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Cubic units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 2

Area of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle of radius 3?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q1/5

2803

QUESTION ID

S76-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q1/5

2803

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle of diameter 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q2/5

2804

QUESTION ID

S76-B2-Q2

TYPE

Short answer

CORRECT ANSWER

25π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A =

A. $2\pi r$

B. πr

C. πr^2

D. πd

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q3/5

2805

QUESTION ID

S76-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. πr^2

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q3/5

True or false?

The base of a cylinder is a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q4/5

2806

QUESTION ID

S76-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q4/5

True or false?

The area of a circle of radius 2 is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q5/5

2807

QUESTION ID

S76-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 3

Cross-sections parallel to the base • Grade 7 • Geometry • 7.G.A.3

True or false?

A slice of a right prism parallel to its base is congruent to the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q1/5

2809

QUESTION ID

S76-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same shape and size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder is sliced parallel to its base. Describe the cross-section.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q2/5

2810

QUESTION ID

S76-B3-Q2

TYPE

Short answer

CORRECT ANSWER

A circle congruent to the base

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q2/5

A rectangular prism sliced parallel to its 3 × 4 base gives a

A. smaller rectangle

B. 3 × 4 rectangle

C. triangle

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q3/5

2811

QUESTION ID

S76-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3 × 4 rectangle

GRADING NOTE Congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why do all horizontal slices of a cylinder have the same area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q4/5

2812

QUESTION ID

S76-B3-Q4

TYPE

Short answer

CORRECT ANSWER

They are all congruent to the base.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q4/5

True or false?

Horizontal slices of a cylinder get smaller toward the top.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q5/5

2813

QUESTION ID

S76-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is a cone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q5/5

SET 76 • HSG-GMD.A.1

FORWARD 1

Cavalieri's principle with different shapes • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder with base area 50 cm² and a square prism with base area 50 cm² both have height 8 cm. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q1/5

2815

QUESTION ID
S76-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Equal: 400 cm³ each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An oblique cylinder has radius 3 and vertical height 7 (its slanted side is 9). Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q2/5

2816

QUESTION ID

S76-F1-Q2

TYPE

Short answer

CORRECT ANSWER

63π

GRADING NOTE Form required: exact value. Use the vertical height, not the slant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q2/5

True or false?

Cavalieri's principle requires equal cross-sectional areas at every height and equal heights.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q3/5

2817

QUESTION ID

S76-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q3/5

Two stacks of 30 identical cards, one straight and one slanted, have

A. different volumes

B. equal volumes

C. equal surface areas

D. no volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q4/5

2818

QUESTION ID

S76-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. equal volumes

GRADING NOTE Same cross-sections.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

State Cavalieri's principle in your own words.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q5/5

2819

QUESTION ID

S76-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Two solids with the same height and equal cross-sectional areas at every level have the same volume.

GRADING NOTE Any equivalent statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSG-GMD.A.1

FORWARD 2

Use the cylinder volume formula to solve problems • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A can has radius 4 cm and height 12 cm. Find its volume to the nearest cubic centimeter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q1/5

2821

QUESTION ID
S76-F2-Q1

TYPE
Short answer

CORRECT ANSWER
603 cm³

GRADING NOTE Rounding to the nearest whole number required. 192π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pipe 2 m long has inner radius 0.1 m. Find the volume of water it holds, to the nearest thousandth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q2/5

2822

QUESTION ID
S76-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.063 m³

GRADING NOTE Rounding to 0.001 required. 0.02π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q2/5

True or false?

Doubling the height of a cylinder multiplies its volume by 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q3/5

2823

QUESTION ID
S76-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Volume is linear in h, so it doubles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Doubling the radius of a cylinder multiplies its volume by

A. 4

B. 8

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q4/5

2824

QUESTION ID

S76-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE r^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q4/5

2824

ANSWER KEY Reduced copy of the question page, with its answer

A cylindrical tank has diameter 6 ft and height 10 ft. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q5/5

2825

QUESTION ID
S76-F2-Q5

TYPE
Short answer

CORRECT ANSWER
90π ft^3

GRADING NOTE Form required: exact value. r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q5/5

SET 77 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the volumes of a pyramid and a cone

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Volume of a prism	**7.G.B.6**
BACKWARD 2	Base area of a pyramid	**6.G.A.1**
BACKWARD 3	The cylinder volume argument	**HSG-GMD.A.1**
FORWARD 1	Cavalieri's principle for pyramids and cones	**HSG-GMD.A.1**
FORWARD 2	Use pyramid and cone volume formulas	**HSG-GMD.A.3**

SET 77 • HSG-GMD.A.1

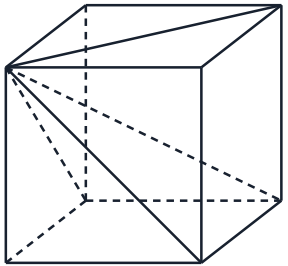
MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A cube is split into three congruent pyramids that share one vertex of the cube as their apex, as shown. Use this to explain why a pyramid's volume is ⅓Bh.

Work space



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q1/5

2828

QUESTION ID

S77-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Each pyramid has the cube's base $B = s^2$ and height $h = s$, and the three fill the cube, so each has volume $\frac{1}{3}s^3 = \frac{1}{3}Bh$.

GRADING NOTE REQUIRED WORK: Compare each pyramid's base and height with the cube's.

Any valid explanation. Essential: three congruent pyramids fill the cube; each has base = a face ($B = s^2$) and height = s; so $V = \frac{1}{3}s^3 = \frac{1}{3}Bh$. Extension to other pyramids by Cavalieri (cross-sections scale the same way) earns full credit but is not required.

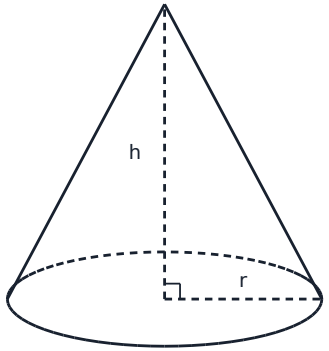
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the volume of a cone is ⅓πr²h, using pyramids.

Work space



Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q2/5

2829

QUESTION ID

S77-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Pyramids with regular n-gon bases inscribed in the circle have V = ⅓Bh; as n grows, B approaches πr² and the pyramids approach the cone, so V = ⅓πr²h.

GRADING NOTE REQUIRED WORK: Think of pyramids whose bases are regular polygons with more and more sides.

Any valid explanation. Essential: pyramid volume ⅓Bh; base polygons approach the circle (B → πr²); the pyramids approach the cone. Alternative: Cavalieri comparison with a pyramid of equal base area and height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone and a cylinder have the same base and the same height. How many cones full of water fill the cylinder?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q3/5

2830

QUESTION ID
S77-M-Q3

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q3/5

A cube with side 6 is split into three congruent pyramids that share one vertex of the cube as their apex. The volume of each pyramid is

A. 36

B. 216

C. 108

D. 72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q4/5

2831

QUESTION ID

S77-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 72

GRADING NOTE 216 ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q4/5

True or false?

A pyramid has half the volume of a prism with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q5/5

2832

QUESTION ID
S77-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE One third.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 1

Volume of a prism • Grade 7 • Geometry • 7.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A prism has base area 36 and height 6. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q1/5

2834

QUESTION ID

S77-B1-Q1

TYPE

Short answer

CORRECT ANSWER

216

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cube has side 6. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q2/5

2835

QUESTION ID

S77-B1-Q2

TYPE

Short answer

CORRECT ANSWER

216

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

V =

A. Bh

B. ⅓Bh

C. 2Bh

D. B + h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q3/5

2836

QUESTION ID

S77-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Bh

GRADING NOTE Prism.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q3/5

True or false?

A cube is a prism.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q4/5

2837

QUESTION ID

S77-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Square base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q4/5

True or false?

A prism's volume depends on the number of its faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q5/5

2838

QUESTION ID
S77-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Base area and height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 2

Base area of a pyramid • Grade 6 • Geometry • 6.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A square base has side 10. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q1/5

2840

QUESTION ID

S77-B2-Q1

TYPE

Short answer

CORRECT ANSWER

100

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangular base has base 6 and height 4. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q2/5

2841

QUESTION ID

S77-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The base of a cone is a

A. triangle

B. circle

C. sector

D. square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q3/5

2842

QUESTION ID

S77-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. circle

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q3/5

True or false?

B in V = ⅓Bh means the area of the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q4/5

2843

QUESTION ID

S77-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q4/5

True or false?

B is the length of a base edge.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q5/5

2844

QUESTION ID

S77-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 3

The cylinder volume argument • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

True or false?

A cylinder's volume is $\pi r^2 h$ because it is a stack of disks of area πr^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q1/5

2846

QUESTION ID

S77-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Set 76.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Cylinder radius 3, height 5: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q2/5

2847

QUESTION ID

S77-B3-Q2

TYPE

Short answer

CORRECT ANSWER

45π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q2/5

A cone fits inside a cylinder with the same base and height. The cone's volume is

A. equal to the cylinder's

B. ⅔ of the cylinder's

C. ½ of the cylinder's

D. ⅓ of the cylinder's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q3/5

2848

QUESTION ID

S77-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. ⅓ of the cylinder's

GRADING NOTE Cone vs cylinder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-B3-Q3**

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder holds 60 cm³. A cone with the same base and height holds

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q4/5

2849

QUESTION ID
S77-B3-Q4

TYPE
Short answer

CORRECT ANSWER
20 cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q4/5

True or false?

A cone is a cylinder with the top cut off flat.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q5/5

2850

QUESTION ID

S77-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It comes to a point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q5/5

SET 77 • HSG-GMD.A.1

FORWARD 1

Cavalieri's principle for pyramids and cones • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid and a cone both have base area 50 cm² and height 9 cm. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q1/5

2852

QUESTION ID

S77-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Equal: 150 cm³ each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An oblique cone has radius 4 and vertical height 6. Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q2/5

2853

QUESTION ID

S77-F1-Q2

TYPE

Short answer

CORRECT ANSWER

32π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q2/5

True or false?

Two pyramids with equal base areas and equal heights have equal volumes even if their apexes are in different positions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q3/5

2854

QUESTION ID

S77-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cavalieri.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why do a pyramid and a cone with equal base areas and heights have equal cross-section areas at every level?

A. Their bases have the same shape

B. Both cross-sections are scaled copies of the base by the same factor

C. They have the same slant height

D. They have the same apex

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q4/5

2855

QUESTION ID
S77-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. Both cross-sections are scaled copies of the base by the same factor

GRADING NOTE Each cross-section at height y is the base scaled by (h − y)/h.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cross-section of a pyramid halfway up has what fraction of the base area?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q5/5

2856

QUESTION ID
S77-F1-Q5

TYPE
Short answer

CORRECT ANSWER
¼

GRADING NOTE Any equivalent form. Scale factor ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSG-GMD.A.1

FORWARD 2

Use pyramid and cone volume formulas • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid has base side 9 m and height 10 m. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q1/5

2858

QUESTION ID
S77-F2-Q1

TYPE
Short answer

CORRECT ANSWER
270 m³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q1/5

2858

ANSWER KEY Reduced copy of the question page, with its answer

A cone has radius 3 in. and height 8 in. Find its volume to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q2/5

2859

QUESTION ID

S77-F2-Q2

TYPE

Short answer

CORRECT ANSWER

75.4 in³

GRADING NOTE Rounding to 0.1 required. 24π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q2/5

True or false?

A cone with the same base and height as a cylinder of volume 90 has volume 45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q3/5

2860

QUESTION ID

S77-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has ⅓ of the volume: 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A pyramid with base area 48 and height 5 has volume

A. 120

B. 53

C. 240

D. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q4/5

2861

QUESTION ID

S77-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 80

GRADING NOTE $\frac{1}{3} \cdot 240$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q4/5

2861

ANSWER KEY Reduced copy of the question page, with its answer

A cone-shaped cup has diameter 6 cm and height 9 cm. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q5/5

2862

QUESTION ID
S77-F2-Q5

TYPE
Short answer

CORRECT ANSWER
27π cm^3

GRADING NOTE Form required: exact value. r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q5/5

SET 78 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

Use volume formulas for cylinders, pyramids, and cones

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	Find a missing side with the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Evaluate a formula and round	**7.EE.B.3**
BACKWARD 4	Which solid uses ⅓?	**HSG-GMD.A.1**
FORWARD 1	Composite and hollow solids	**HSG-GMD.A.3**
FORWARD 2	Mass from volume and density	**HSG-MG.A.2**

SET 78 • HSG-GMD.A.3

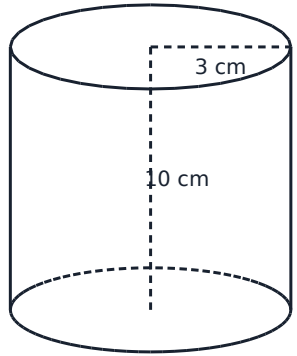
MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder has radius 3 cm and height 10 cm. Find its volume, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q1/5

2865

QUESTION ID

S78-M-Q1

TYPE

Short answer

CORRECT ANSWER

90π cm³ ≈ 282.7 cm³

GRADING NOTE Form required: both the exact value and rounding to 0.1, with units.

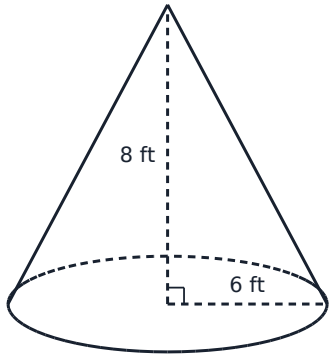
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has radius 6 ft and height 8 ft. Find its volume, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q2/5

2866

QUESTION ID

S78-M-Q2

TYPE

Short answer

CORRECT ANSWER

96π ft³ ≈ 301.6 ft³

GRADING NOTE Form required: both, with units. ⅓π · 36 · 8.

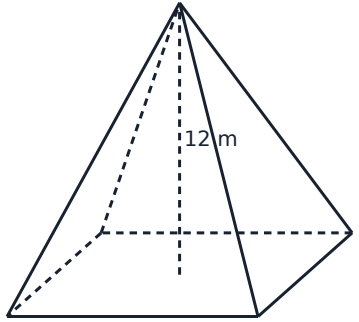
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid has base side 10 m and height 12 m. Find its volume.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q3/5

2867

QUESTION ID

S78-M-Q3

TYPE

Short answer

CORRECT ANSWER

400 m³

GRADING NOTE Any equivalent form, with units. ⅓ · 100 · 12.

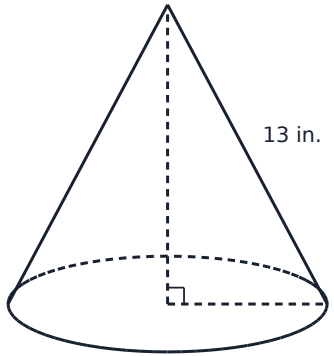
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has diameter 10 in. and slant height 13 in. Find its volume, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q4/5

2868

QUESTION ID
S78-M-Q4

TYPE
Short answer

CORRECT ANSWER
100π in³ ≈ 314.2 in³

GRADING NOTE Form required: both, with units. r = 5; height √(169 − 25) = 12; ⅓π · 25 · 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder and a cone have the same radius and height. The cylinder holds 600 mL. The cone holds

A. 300 mL

B. 200 mL

C. 600 mL

D. 1,800 mL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q5/5

2869

QUESTION ID

S78-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 200 mL

GRADING NOTE $\frac{1}{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle with radius 6?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q1/5

2871

QUESTION ID

S78-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle with diameter 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q2/5

2872

QUESTION ID

S78-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q2/5

A base of radius 3 has area

A. 9

B. 9π

C. 3π

D. 6π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q3/5

2873

QUESTION ID

S78-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9π

GRADING NOTE πr^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q3/5

True or false?

Squaring the radius comes before multiplying by π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q4/5

2874

QUESTION ID

S78-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Order of operations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

πr^2 with $r = 3$ is $(3\pi)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q5/5

2875

QUESTION ID

S78-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9π, not $9\pi^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 2

Find a missing side with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has hypotenuse 13 and one leg 5. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q1/5

2877

QUESTION ID

S78-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has radius 6 and slant height 10. Height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q2/5

2878

QUESTION ID

S78-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q2/5

Radius, height, and slant height of a cone form a

A. square

B. right triangle

C. equilateral triangle

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q3/5

2879

QUESTION ID

S78-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. right triangle

GRADING NOTE Height ⊥ base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q3/5

True or false?

The slant height is longer than the height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q4/5

2880

QUESTION ID

S78-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q4/5

True or false?

The slant height can be used as the height in $V = \frac{1}{3}\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q5/5

2881

QUESTION ID

S78-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use the perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 3

Evaluate a formula and round • Grade 7 • Expressions & Equations • 7.EE.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate ⅓ · 36 · 8.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q1/5

2883

QUESTION ID

S78-B3-Q1

TYPE

Short answer

CORRECT ANSWER

96

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Round 90π to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q2/5

2884

QUESTION ID

S78-B3-Q2

TYPE

Short answer

CORRECT ANSWER

282.7

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$96\pi \approx$

A. 301.6

B. 301.5

C. 3,016

D. 96.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q3/5

2885

QUESTION ID

S78-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 301.6

GRADING NOTE Use the π key.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Using 3.14 for π may change the last digit of a rounded answer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q4/5

2886

QUESTION ID

S78-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Use the π key.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\frac{1}{3} \cdot 100 \cdot 12 = 1{,}200$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q5/5

2887

QUESTION ID

S78-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 4

Which solid uses ⅓? • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

True or false?

Pyramids and cones use V = ⅓Bh.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q1/5

2889

QUESTION ID

S78-B4-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pointed solids.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which has the formula V = Bh: a cone or a cylinder?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q2/5

2890

QUESTION ID

S78-B4-Q2

TYPE

Short answer

CORRECT ANSWER

Cylinder

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$V = \frac{1}{3}\pi r^2 h$ is the volume of a

A. cylinder

B. cone

C. prism

D. sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q3/5

2891

QUESTION ID

S78-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. cone

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why does a cone have ⅓ of the matching cylinder's volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q4/5

2892

QUESTION ID

S78-B4-Q4

TYPE

Short answer

CORRECT ANSWER

Three cones fill the cylinder (the informal argument).

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A pyramid uses V = Bh.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q5/5

2893

QUESTION ID

S78-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ⅓Bh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q5/5

2893

SET 78 • HSG-GMD.A.3

FORWARD 1

Composite and hollow solids • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A silo is a cylinder (radius 4 m, height 10 m) topped by a cone (radius 4 m, height 3 m). Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q1/5

2895

QUESTION ID
S78-F1-Q1

TYPE
Short answer

CORRECT ANSWER
176π m^3

GRADING NOTE Form required: exact value. $160\pi + 16\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pipe is a cylinder of length 20 cm with outer radius 5 cm and inner radius 4 cm. Find the volume of the material, exact.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q2/5

2896

QUESTION ID
S78-F1-Q2

TYPE
Short answer

CORRECT ANSWER
180π cm^3

GRADING NOTE Form required: exact value. $\pi(25 - 16)(20)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q2/5

True or false?

The volume of a composite solid is the sum of the volumes of its non-overlapping parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q3/5

2897

QUESTION ID

S78-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Additivity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder (V = 300) has a cone with the same base and height carved out. Remaining volume?

A. 300

B. 200

C. 0

D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q4/5

2898

QUESTION ID

S78-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 200

GRADING NOTE 300 − 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A pencil tip is a cone of radius 0.4 cm and height 1.5 cm on a cylinder of radius 0.4 cm and length 15 cm. Find the total volume to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q5/5

2899

QUESTION ID
S78-F1-Q5

TYPE
Short answer

CORRECT ANSWER
7.8 cm³

GRADING NOTE Rounding to 0.1 required. 2.4π + 0.08π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q5/5

SET 78 • HSG-GMD.A.3

FORWARD 2

Mass from volume and density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A steel cylinder has radius 2 cm and height 5 cm. Steel has density 7.85 g/cm³. Find the mass to the nearest gram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q1/5

2901

QUESTION ID
S78-F2-Q1

TYPE
Short answer

CORRECT ANSWER
493 g

GRADING NOTE Rounding to the nearest gram required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone of ice cream (radius 3 cm, height 10 cm) has density 0.55 g/cm³. Find the mass to the nearest gram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q2/5

2902

QUESTION ID
S78-F2-Q2

TYPE
Short answer

CORRECT ANSWER
52 g

GRADING NOTE Rounding to the nearest gram required. 30π cm³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Mass = volume ÷ density.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q3/5

2903

QUESTION ID

S78-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Mass = density × volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q3/5

A pyramid of volume 400 cm³ is made of material with density 2 g/cm³. Mass?

A. 200 g

B. 402 g

C. 1,600 g

D. 800 g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q4/5

2904

QUESTION ID

S78-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 800 g

GRADING NOTE 2 × 400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A gold cube has volume 8 cm³ and mass 154.4 g. Find the density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q5/5

2905

QUESTION ID
S78-F2-Q5

TYPE
Short answer

CORRECT ANSWER
19.3 g/cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q5/5

SET 79 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

Use the volume formula for spheres, including composite solids

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Cubes of numbers	**6.EE.A.1**
BACKWARD 2	Radius from diameter	**7.G.B.4**
BACKWARD 3	Multiply fractions by numbers	**7.NS.A.2**
BACKWARD 4	Cylinder and cone volumes	**HSG-GMD.A.3**
FORWARD 1	Scaling spheres	**HSG-GMD.A.3**
FORWARD 2	Model real objects as spheres	**HSG-MG.A.1**

SET 79 • HSG-GMD.A.3

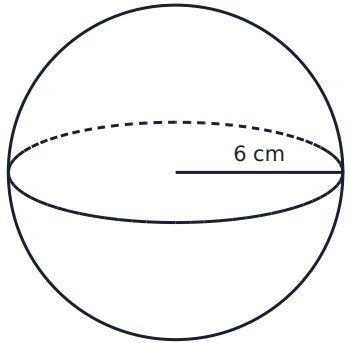
MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find the volume of a sphere with radius 6 cm, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q1/5

2908

QUESTION ID
S79-M-Q1

TYPE
Short answer

CORRECT ANSWER
288π cm³ ≈ 904.8 cm³

GRADING NOTE Form required: both, with units. 4/3 π · 216.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the volume of a ball with diameter 10 in., exact and to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q2/5

2909

QUESTION ID

S79-M-Q2

TYPE

Short answer

CORRECT ANSWER

$500\pi/3$ in^3 ≈ 523.6 in^3

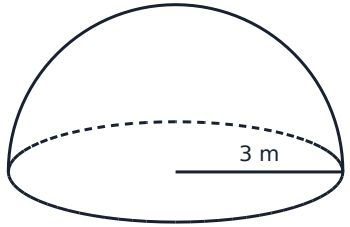
GRADING NOTE Form required: both, with units. r = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the volume of a hemisphere with radius 3 m, exact and to the nearest tenth.



Volume: _____

Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q3/5

2910

QUESTION ID

S79-M-Q3

TYPE

Short answer

CORRECT ANSWER

18π m³ ≈ 56.5 m³

GRADING NOTE Form required: both, with units. ½ · 4/3 π · 27.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A capsule is a cylinder of radius 3 cm and length 10 cm with a hemisphere on each end. Find its volume, exact and to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q4/5

2911

QUESTION ID

S79-M-Q4

TYPE

Short answer

CORRECT ANSWER

126π cm³ ≈ 395.8 cm³

GRADING NOTE Form required: both, with units. 90π + 36π (two hemispheres make one sphere).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q4/5

An ice cream cone (radius 2, height 9) is topped by a hemisphere of radius 2. The total exact volume is

A. 52π/3

B. 28π

C. 12π

D. 44π/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q5/5

2912

QUESTION ID

S79-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 52π/3

GRADING NOTE 12π + 16π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 1

Cubes of numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

$6^3 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q1/5

2914

QUESTION ID

S79-B1-Q1

TYPE

Short answer

CORRECT ANSWER

216

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$5^3 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q2/5

2915

QUESTION ID

S79-B1-Q2

TYPE

Short answer

CORRECT ANSWER

125

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$3^3 =$

A. 9

B. 81

C. 6

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q3/5

2916

QUESTION ID

S79-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 27

GRADING NOTE $3 \cdot 3 \cdot 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$2^3 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q4/5

2917

QUESTION ID

S79-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cube.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$4^3 = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q5/5

2918

QUESTION ID

S79-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 2

Radius from diameter • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A ball has diameter 10. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q1/5

2920

QUESTION ID

S79-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q1/5

True or false?

Using the diameter in $4/3\ \pi r^3$ makes the answer 8 times too large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q2/5

2921

QUESTION ID

S79-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere with diameter 12 has r^3 =

A. 216

B. 144

C. 36

D. 1,728

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q3/5

2922

QUESTION ID

S79-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 216

GRADING NOTE 6^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q3/5

2922

ANSWER KEY Reduced copy of the question page, with its answer

A sphere has radius 2.5. Diameter?

Diameter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q4/5

2923

QUESTION ID

S79-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q4/5

True or false?

The radius of a sphere is twice its diameter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q5/5

2924

QUESTION ID

S79-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 3

Multiply fractions by numbers • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$4/3 \cdot 216 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q1/5

2926

QUESTION ID

S79-B3-Q1

TYPE

Short answer

CORRECT ANSWER

288

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

½ · 4/3 · 27 =

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q2/5

2927

QUESTION ID

S79-B3-Q2

TYPE

Short answer

CORRECT ANSWER

18

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$4/3 \cdot 125 =$

A. 375/4

B. 500/3

C. 125/3

D. 500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q3/5

2928

QUESTION ID

S79-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 500/3

GRADING NOTE Numerators.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$4/3 \cdot 27 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q4/5

2929

GRADING NOTE $27 \div 3 \times 4$.

QUESTION ID

S79-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$4/3 \cdot 6 = 24$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q5/5

2930

QUESTION ID

S79-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 4

Cylinder and cone volumes • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Cylinder radius 3, length 10: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q1/5

2932

QUESTION ID

S79-B4-Q1

TYPE

Short answer

CORRECT ANSWER

90π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Cone radius 2, height 9: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q2/5

2933

QUESTION ID

S79-B4-Q2

TYPE

Short answer

CORRECT ANSWER

12π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder and a cone with radius 3 and height 4 have volumes

A. 36π and 12π

B. 12π and 36π

C. 12π and 4π

D. 36π and 36π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q3/5

2934

QUESTION ID

S79-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 36π and 12π

GRADING NOTE ⅓.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q3/5

True or false?

Two hemispheres of radius r make one sphere of radius r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q4/5

2935

QUESTION ID

S79-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Halves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q4/5

True or false?

A composite volume is the product of the parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q5/5

2936

QUESTION ID

S79-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q5/5

SET 79 • HSG-GMD.A.3

FORWARD 1

Scaling spheres • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

The radius of a sphere is doubled. By what factor is the volume multiplied?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q1/5

2938

QUESTION ID

S79-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. 2^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere fits exactly inside a cube with side 6. Find the exact volume of the space in the cube outside the sphere.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q2/5

2939

QUESTION ID
S79-F1-Q2

TYPE
Short answer

CORRECT ANSWER
216 − 36π

GRADING NOTE Form required: exact value. r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q2/5

2939

True or false?

A sphere of radius 3 has 9 times the volume of a sphere of radius 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q3/5

2940

QUESTION ID

S79-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volume scales by $3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A balloon's radius grows from 10 cm to 15 cm. Its volume is multiplied by

A. 1.5

B. 2.25

C. 5

D. 3.375

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q4/5

2941

QUESTION ID

S79-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3.375

GRADING NOTE 1.5^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere and a cylinder have the same radius r, and the cylinder's height is 2r. What fraction of the cylinder's volume is the sphere's?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q5/5

2942

QUESTION ID

S79-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{3}$

GRADING NOTE Any equivalent form. $(4/3\ \pi r^3)/(2\pi r^3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q5/5

SET 79 • HSG-GMD.A.3

FORWARD 2

Model real objects as spheres • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Model Earth as a sphere with radius 6,371 km. Estimate its volume in cubic kilometers, in scientific notation to 3 significant figures.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q1/5

2944

QUESTION ID

S79-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.08×10^{12} km^3

GRADING NOTE Rounding to 3 significant figures required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q1/5

A basketball has circumference 75 cm. Model it as a sphere and find its volume to the nearest 100 cm^3.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q2/5

2945

QUESTION ID

S79-F2-Q2

TYPE

Short answer

CORRECT ANSWER

7100 cm^3

GRADING NOTE Rounding to the nearest 100 required. r = 75/(2π) ≈ 11.94 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q2/5

True or false?

Modeling an orange as a sphere ignores its small bumps.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q3/5

2946

QUESTION ID

S79-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Modeling assumption.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q3/5

2946

ANSWER KEY Reduced copy of the question page, with its answer

A scoop of ice cream is modeled as a sphere of radius 3 cm. Its volume is about

A. 28 cm³

B. 113 cm³

C. 339 cm³

D. 36 cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q4/5

2947

QUESTION ID
S79-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 113 cm³

GRADING NOTE 36π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give one reason a sphere is a good model for a ball bearing and one limitation of the model for a pumpkin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q5/5

2948

QUESTION ID
S79-F2-Q5

TYPE
Short answer

CORRECT ANSWER
A ball bearing is nearly perfectly round; a pumpkin has ridges, a flattened shape, and a stem, so the model is only approximate.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q5/5

Solve for an unknown dimension from a volume

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Solve a one-step equation with a coefficient	**HSA-REI.B.3**
BACKWARD 2	Square roots and cube roots	**8.EE.A.2**
BACKWARD 3	Rearrange a formula	**HSA-CED.A.4**
FORWARD 1	Find dimensions to meet a target volume	**HSG-GMD.A.3**
FORWARD 2	Design a container that meets constraints	**HSG-MG.A.3**

SET 80 • HSG-GMD.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder has volume 360π cm³ and radius 6 cm. Find its height.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q1/5

2951

QUESTION ID

S80-M-Q1

TYPE

Short answer

CORRECT ANSWER

10 cm

GRADING NOTE Any equivalent form, with units. $36\pi h = 360\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has volume 96π in^3 and height 8 in. Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q2/5

2952

QUESTION ID
S80-M-Q2

TYPE
Short answer

CORRECT ANSWER
6 in.

GRADING NOTE Any equivalent form, with units. $\frac{1}{3}\pi r^2 \cdot 8 = 96\pi \rightarrow r^2 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere has volume 36π m³. Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q3/5

2953

QUESTION ID
S80-M-Q3

TYPE
Short answer

CORRECT ANSWER
3 m

GRADING NOTE Any equivalent form, with units. $r^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q3/5

2953

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid has volume 256 ft³ and height 12 ft. Find the side of its base.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q4/5

2954

QUESTION ID
S80-M-Q4

TYPE
Short answer

CORRECT ANSWER
8 ft

GRADING NOTE Any equivalent form, with units. $\frac{1}{3}s^2 \cdot 12 = 256 \rightarrow s^2 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder holds 500 cm³ and has height 10 cm. Its radius is closest to

A. 7.1 cm

B. 4.0 cm

C. 2.0 cm

D. 15.9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q5/5

2955

QUESTION ID

S80-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 4.0 cm

GRADING NOTE $r = \sqrt{(50/\pi)} \approx 3.99$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q5/5

SET 80 • HSG-GMD.A.3

BACKWARD 1

Solve a one-step equation with a coefficient • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 36πh = 360π.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q1/5

2957

QUESTION ID

S80-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B1-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ⅓x = 12.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q2/5

2958

QUESTION ID

S80-B1-Q2

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B1-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 4x = 108.

A. 25

B. 104

C. 432

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q3/5

2959

QUESTION ID

S80-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 27

GRADING NOTE Divide by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B1-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q3/5

True or false?

To solve $36\pi h = 360\pi$, dividing both sides by π is allowed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q4/5

2960

QUESTION ID

S80-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\pi \neq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

To solve ⅓x = 12, divide by 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q5/5

2961

QUESTION ID

S80-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Multiply by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q5/5

SET 80 • HSG-GMD.A.3

BACKWARD 2

Square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve $r^2 = 36$ for $r > 0$.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q1/5

2963

QUESTION ID

S80-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $r^3 = 27$.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q2/5

2964

QUESTION ID

S80-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sqrt[3]{125} =$

A. 25

B. 5

C. 41.7

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q3/5

2965

QUESTION ID

S80-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

GRADING NOTE $5^3 = 125$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{64} = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q4/5

2966

QUESTION ID

S80-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $8^2 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $r^3 = 8$, then $r = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q5/5

2967

QUESTION ID

S80-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSG-GMD.A.3

BACKWARD 3

Rearrange a formula • High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $V = \pi r^2 h$ for h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q1/5

2969

QUESTION ID

S80-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$V/(\pi r^2)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B3-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve V = ⅓Bh for B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q2

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q2/5

2970

QUESTION ID

S80-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3V/h

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B3-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$V = 4/3\ \pi r^3$ solved for r^3 is

A. $4V/(3\pi)$

B. $V/(4\pi)$

C. $3V/(4\pi)$

D. $3\pi V/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q3

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q3/5

2971

QUESTION ID
S80-B3-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. 3V/(4π)

GRADING NOTE Multiply by 3/(4π).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q3

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q3/5

True or false?

Solving $V = \pi r^2 h$ for r needs a square root.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q4/5

2972

QUESTION ID
S80-B3-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $r = \sqrt{(V/(\pi h))}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q4/5

True or false?

Solving V = Bh for h gives h = VB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q5/5

2973

QUESTION ID
S80-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE V/B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q5/5

SET 80 • HSG-GMD.A.3

FORWARD 1

Find dimensions to meet a target volume • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A drink can must hold 355 cm³ and have radius 3.3 cm. Find the height to the nearest tenth of a centimeter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q1/5

2975

QUESTION ID
S80-F1-Q1

TYPE
Short answer

CORRECT ANSWER
10.4 cm

GRADING NOTE Rounding to 0.1 required. $h = 355/(\pi \cdot 3.3^2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A spherical tank must hold 1,000 m³. Find its radius to the nearest tenth of a meter.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q2/5

2976

QUESTION ID
S80-F1-Q2

TYPE
Short answer

CORRECT ANSWER
6.2 m

GRADING NOTE Rounding to 0.1 required. $r = \sqrt[3]{(750/\pi)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q2/5

True or false?

To double the volume of a cylinder while keeping its radius, double its height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q3/5

2977

QUESTION ID

S80-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Linear in h.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has the same volume as a cylinder of equal radius. The cone's height is

A. 9 times the cylinder's

B. ⅓ of the cylinder's

C. 3 times the cylinder's

D. equal to the cylinder's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q4/5

2978

QUESTION ID

S80-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3 times the cylinder's

GRADING NOTE $\frac{1}{3}\pi r^2 h_1 = \pi r^2 h_2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cube has the same volume as a sphere of radius 3. Find the cube's edge to the nearest tenth.

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q5/5

2979

QUESTION ID

S80-F1-Q5

TYPE

Short answer

CORRECT ANSWER

4.8

GRADING NOTE Rounding to 0.1 required. $\sqrt[3]{(36\pi)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q5/5

SET 80 • HSG-GMD.A.3

FORWARD 2

Design a container that meets constraints • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A cylindrical vase must hold 2,000 cm³ and be 20 cm tall. Its radius must be at least 5 cm to fit a bouquet. Does the design work? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q1/5

2981

QUESTION ID

S80-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: r = √(2000/(20π)) ≈ 5.6 cm, which is at least 5 cm.

GRADING NOTE Form required: the conclusion with the computed radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A conical paper cup must hold 150 cm³ and have radius 3.5 cm. Find the height to the nearest tenth.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q2/5

2982

QUESTION ID
S80-F2-Q2

TYPE
Short answer

CORRECT ANSWER
11.7 cm

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q2/5

True or false?

A container's volume requirement fixes each of its dimensions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q3/5

2983

QUESTION ID

S80-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It fixes only a relationship: many (r, h) pairs give the same volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q3/5

A cylinder must hold 100π cm^3. Which (r, h) works?

A. (5, 4)

B. (5, 20)

C. (4, 5)

D. (10, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q4/5

2984

QUESTION ID

S80-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (5, 4)

GRADING NOTE $25 \cdot 4 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give two different cylinders (radius and height) that each hold 64π cm^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q5/5

2985

GRADING NOTE Any two (r, h) pairs with $r^2h = 64$.

QUESTION ID

S80-F2-Q5

TYPE

Short answer

CORRECT ANSWER

For example r = 4, h = 4 and r = 2, h = 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q5/5

SET 81 • HSG-GMD.B.4 • GEOMETRIC MEASUREMENT AND DIMENSION

Identify cross-sections of three-dimensional objects

MAIN	Geometry main question	**HSG-GMD.B.4**
BACKWARD 1	Cross-sections of right rectangular prisms and pyramids	**7.G.A.3**
BACKWARD 2	Properties of polygons	**5.G.B.4**
BACKWARD 3	Parts of solids	**6.G.A.4**
FORWARD 1	Areas of cross-sections	**HSG-GMD.B.4**
FORWARD 2	Use cross-section areas to compare volumes (Cavalieri)	**HSG-GMD.A.2**

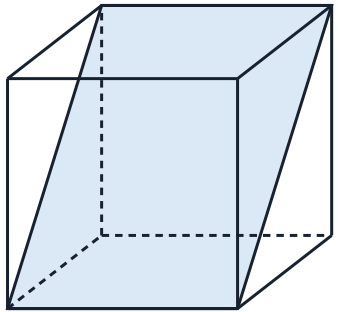
SET 81 • HSG-GMD.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A plane cuts the cube as shown. Name the shape of the cross-section.



Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q1/5

2988

QUESTION ID

S81-M-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle (not a square)

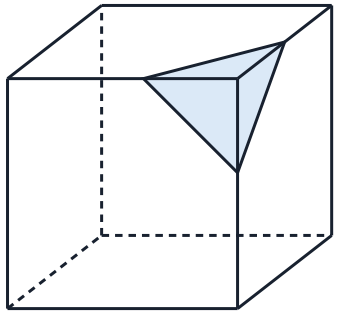
GRADING NOTE Any equivalent answer. Two sides are cube edges (s) and two are face diagonals (s√2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane cuts off one corner of the cube as shown. Name the shape of the cross-section.



Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q2/5

2989

QUESTION ID

S81-M-Q2

TYPE

Short answer

CORRECT ANSWER

A triangle

GRADING NOTE Any equivalent answer. (Equilateral if the three cut points are equally far from the corner.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A right circular cone is cut by a vertical plane through its apex. The cross-section is

A. a circle

B. a rectangle

C. a triangle (isosceles)

D. an ellipse

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q3/5

2990

QUESTION ID

S81-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. a triangle (isosceles)

GRADING NOTE Two slant edges and a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Name the cross-section of a cylinder cut by a plane (a) parallel to its base, (b) perpendicular to its base through its axis, and (c) slanted, cutting the curved side only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q4/5

2991

QUESTION ID

S81-M-Q4

TYPE

Short answer

CORRECT ANSWER

(a) circle (b) rectangle (c) ellipse

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q4/5

True or false?

Every plane section of a sphere that cuts through it is a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q5/5

2992

QUESTION ID

S81-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Points of the sphere in the plane are equidistant from the foot of the perpendicular from the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 1

Cross-sections of right rectangular prisms and pyramids • Grade 7 • Geometry • 7.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A rectangular prism is cut parallel to its base. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q1/5

2994

QUESTION ID

S81-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle congruent to the base

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid is cut parallel to its base. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q2/5

2995

QUESTION ID

S81-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A smaller square

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q2/5

A square pyramid cut vertically through its apex and two opposite base edges' midpoints gives a

A. trapezoid

B. square

C. circle

D. triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q3/5

2996

QUESTION ID

S81-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. triangle

GRADING NOTE Through the apex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q3/5

True or false?

A vertical cut of a rectangular prism parallel to a side face gives a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q4/5

2997

QUESTION ID

S81-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Congruent to the face.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q4/5

True or false?

A cut of a cube parallel to a face can be a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q5/5

2998

QUESTION ID

S81-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 2

Properties of polygons • Grade 5 • Geometry • 5.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A quadrilateral with four right angles and unequal adjacent sides is a

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q1/5

3000

QUESTION ID

S81-B2-Q1

TYPE

Short answer

CORRECT ANSWER

rectangle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q1/5

3000

True or false?

Every square is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q2/5

3001

QUESTION ID

S81-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Hierarchy.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A polygon with six sides is a

A. hexagon

B. pentagon

C. octagon

D. heptagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q3/5

3002

QUESTION ID

S81-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. hexagon

GRADING NOTE Name.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle with two equal sides is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q4/5

3003

QUESTION ID

S81-B2-Q4

TYPE

Short answer

CORRECT ANSWER

isosceles

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q4/5

3003

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A trapezoid has no parallel sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q5/5

3004

QUESTION ID

S81-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At least one pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 3

Parts of solids • Grade 6 • Geometry • 6.G.A.4

ANSWER KEY Reduced copy of the question page, with its answer

How many faces does a cube have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q1/5

3006

QUESTION ID

S81-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the shape of the base of a cone?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q2/5

3007

QUESTION ID

S81-B3-Q2

TYPE

Short answer

CORRECT ANSWER

A circle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q2/5

How many edges meet at each vertex of a cube?

A. 6

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q3/5

3008

QUESTION ID

S81-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE Corner.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q3/5

True or false?

A cylinder has two congruent circular bases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q4/5

3009

QUESTION ID

S81-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q4/5

True or false?

A sphere has one edge.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q5/5

3010

QUESTION ID

S81-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No edges.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q5/5

SET 81 • HSG-GMD.B.4

FORWARD 1

Areas of cross-sections • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A cube has edge 4. Find the area of the diagonal cross-section through two opposite edges.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q1/5

3012

QUESTION ID
S81-F1-Q1

TYPE
Short answer

CORRECT ANSWER
16√2

GRADING NOTE Form required: exact value. 4 × 4√2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has radius 6 and height 12. Find the area of the cross-section parallel to the base at height 4 (measured up from the base).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q2/5

3013

QUESTION ID

S81-F1-Q2

TYPE

Short answer

CORRECT ANSWER

16π

GRADING NOTE Form required: exact value. Radius at that level = 6 · (12 − 4)/12 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q2/5

True or false?

A sphere of radius 5 cut 3 units from its center gives a circle of radius 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q3/5

3014

QUESTION ID

S81-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3-4-5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q3/5

A cube can have a cross-section that is a regular

A. octagon

B. pentagon

C. heptagon

D. hexagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q4/5

3015

QUESTION ID

S81-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. hexagon

GRADING NOTE Plane through the midpoints of six edges.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F1-Q4**

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid has base side 10 and height 15. Find the area of the cross-section parallel to the base 5 units below the apex.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q5/5

3016

QUESTION ID
S81-F1-Q5

TYPE
Short answer

CORRECT ANSWER
100/9 (about 11.1)

GRADING NOTE Form required: exact value or rounding to 0.1. Side 10 · 5/15 = 10/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSG-GMD.B.4

FORWARD 2

Use cross-section areas to compare volumes (Cavalieri) • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Two solids have height 10 and every horizontal cross-section of each has area 7. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q1/5

3018

QUESTION ID

S81-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Equal: 70 each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A hemisphere has radius 5. Find the area of its cross-section 3 units above the base.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q2/5

3019

QUESTION ID

S81-F2-Q2

TYPE

Short answer

CORRECT ANSWER

16π

GRADING NOTE Form required: exact value. Radius 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q2/5

True or false?

Cavalieri's principle can compare two solids of different heights.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q3/5

3020

QUESTION ID

S81-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solids must have equal heights.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q3/5

At height y, a hemisphere of radius r has cross-section area

A. $\pi(r^2 - y^2)$

B. $\pi(r - y)^2$

C. πr^2

D. πy^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q4/5

3021

QUESTION ID

S81-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $\pi(r^2 - y^2)$

GRADING NOTE Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why cross-section areas are useful for finding volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q5/5

3022

GRADING NOTE Any reasonable answer.

QUESTION ID

S81-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Volume can be built from thin slices; equal slice areas at each level give equal volumes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q5/5

SET 82 • HSG-GMD.B.4 • GEOMETRIC MEASUREMENT AND DIMENSION

Identify three-dimensional objects generated by rotations of two-dimensional objects

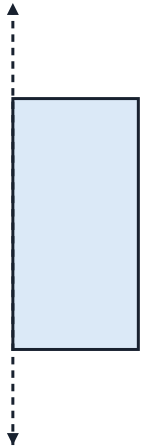
MAIN	Geometry main question	**HSG-GMD.B.4**
BACKWARD 1	Rotation of a point about a center	**HSG-CO.A.4**
BACKWARD 2	Cross-sections perpendicular to the axis	**HSG-GMD.B.4**
BACKWARD 3	Names of solids	**6.G.A.4**
FORWARD 1	Volumes of solids of rotation	**HSG-GMD.B.4**
FORWARD 2	Design an object as a solid of rotation	**HSG-MG.A.3**

SET 82 • HSG-GMD.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

The rectangle is rotated 360° about the dashed line. Name the solid formed.



Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q1/5

3025

QUESTION ID

S82-M-Q1

TYPE

Short answer

CORRECT ANSWER

A cylinder

GRADING NOTE Any equivalent answer.

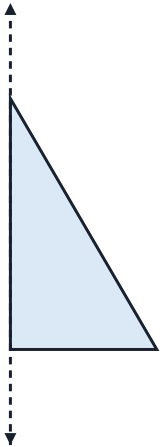
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The right triangle is rotated 360° about the dashed line. Name the solid formed.

Solid: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q2/5

3026

QUESTION ID

S82-M-Q2

TYPE

Short answer

CORRECT ANSWER

A cone

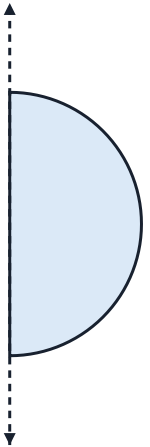
GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The semicircle is rotated 360° about the dashed line. Name the solid formed.



Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q3/5

3027

QUESTION ID

S82-M-Q3

TYPE

Short answer

CORRECT ANSWER

A sphere

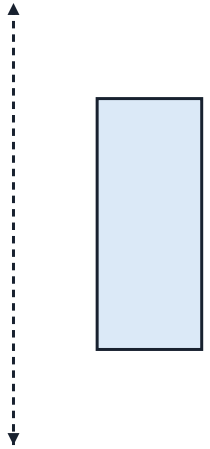
GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The rectangle is rotated 360° about the dashed line, which does not touch it. Describe the solid formed.



Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q4/5

3028

QUESTION ID

S82-M-Q4

TYPE

Short answer

CORRECT ANSWER

A hollow cylinder (a cylinder with a cylindrical hole through the middle, like a pipe)

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q4/5

3028

A right triangle is rotated about its hypotenuse. The solid formed is

A. a sphere

B. one cone

C. two cones joined at their bases

D. a cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q5/5

3029

QUESTION ID

S82-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. two cones joined at their bases

GRADING NOTE Each leg sweeps a cone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 1

Rotation of a point about a center • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

True or false?

Rotating a point 360° about a center traces a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q1/5

3031

QUESTION ID

S82-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A point 3 cm from an axis is rotated about the axis. What does it trace?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q2/5

3032

QUESTION ID

S82-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A circle of radius 3 cm

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A point ON the axis of rotation

A. traces a line

B. stays fixed

C. traces a circle

D. disappears

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q3/5

3033

QUESTION ID

S82-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. stays fixed

GRADING NOTE Distance 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q3/5

A segment parallel to the axis, 2 units away, is rotated about it. What surface does it trace?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q4/5

3034

QUESTION ID

S82-B1-Q4

TYPE

Short answer

CORRECT ANSWER

The curved side of a cylinder

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q4/5

True or false?

Points farther from the axis trace smaller circles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q5/5

3035

QUESTION ID

S82-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Larger circles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 2

Cross-sections perpendicular to the axis • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

True or false?

A solid of rotation cut perpendicular to its axis has a circular cross-section (or a ring).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q1/5

3037

QUESTION ID

S82-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every point traces a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone is cut perpendicular to its axis. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q2/5

3038

QUESTION ID

S82-B2-Q2

TYPE

Short answer

CORRECT ANSWER

A circle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A pipe cut perpendicular to its axis gives a

A. rectangle

B. ring (annulus)

C. circle

D. triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q3/5

3039

QUESTION ID

S82-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. ring (annulus)

GRADING NOTE Hole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere is cut through its center. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q4/5

3040

QUESTION ID

S82-B2-Q4

TYPE

Short answer

CORRECT ANSWER

A circle (a great circle)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q4/5

True or false?

A cylinder cut perpendicular to its axis gives a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q5/5

3041

QUESTION ID

S82-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 3

Names of solids • Grade 6 • Geometry • 6.G.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Name the solid with two parallel circular bases and a curved side.

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q1/5

3043

QUESTION ID

S82-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Cylinder

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Name the solid with one circular base and an apex.

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q2/5

3044

QUESTION ID

S82-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Cone

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q2/5

All points equidistant from a center in space form a

A. sphere

B. cone

C. circle

D. cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q3/5

3045

QUESTION ID

S82-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. sphere

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q3/5

True or false?

A cone has one flat face.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q4/5

3046

QUESTION ID

S82-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Its base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q4/5

True or false?

A sphere has a flat base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q5/5

3047

QUESTION ID

S82-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q5/5

SET 82 • HSG-GMD.B.4

FORWARD 1

Volumes of solids of rotation • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A 3-by-5 rectangle is rotated about its 5-unit side. Find the exact volume of the solid.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q1/5

3049

QUESTION ID
S82-F1-Q1

TYPE
Short answer

CORRECT ANSWER
45π

GRADING NOTE Form required: exact value. r = 3, h = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 3-by-5 rectangle is rotated about its 3-unit side. Find the exact volume of the solid.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q2/5

3050

QUESTION ID
S82-F1-Q2

TYPE
Short answer

CORRECT ANSWER
75π

GRADING NOTE Form required: exact value. r = 5, h = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q2/5

3050

True or false?

Rotating one rectangle about two different sides can give solids with different volumes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q3/5

3051

QUESTION ID

S82-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A 3-by-5 rectangle gives 45π about its 5-unit side and 75π about its 3-unit side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle with legs 3 (on the axis) and 4 is rotated about the leg of length 3. Volume?

A. 48π

B. 9π

C. 12π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q4/5

3052

QUESTION ID

S82-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 16π

GRADING NOTE r = 4, h = 3: ⅓π · 16 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A semicircle of radius 3 is rotated about its diameter. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q5/5

3053

QUESTION ID
S82-F1-Q5

TYPE
Short answer

CORRECT ANSWER
36π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q5/5

SET 82 • HSG-GMD.B.4

FORWARD 2

Design an object as a solid of rotation • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A lathe turns a 2 cm by 10 cm rectangle about its 10 cm side to make a wooden dowel. Find the exact volume of wood.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q1/5

3055

QUESTION ID

S82-F2-Q1

TYPE

Short answer

CORRECT ANSWER

40π cm^3

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q1/5

3055

A designer wants a cone-shaped lampshade by rotating a right triangle about a 30 cm leg. What other leg length gives a base of diameter 40 cm?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q2/5

3056

QUESTION ID

S82-F2-Q2

TYPE

Short answer

CORRECT ANSWER

20 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q2/5

True or false?

A hemispherical bowl can be modeled by rotating a right triangle about one of its legs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q3/5

3057

QUESTION ID

S82-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That gives a cone; rotate a quarter-circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q3/5

To make a ring-shaped washer by rotation, rotate a rectangle about a line that

A. is a diagonal

B. contains one side

C. does not touch the rectangle

D. passes through its center

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q4/5

3058

QUESTION ID

S82-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. does not touch the rectangle

GRADING NOTE Gap gives the hole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A vase is made by rotating a rectangle 6 cm wide and 20 cm tall about a line 2 cm from its long side (outside the rectangle). Find the exact volume of material.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q5/5

3059

QUESTION ID
S82-F2-Q5

TYPE
Short answer

CORRECT ANSWER
1,200π cm³

GRADING NOTE Form required: exact value. Outer radius 8, inner radius 2: π(64 − 4)(20).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q5/5

SET 83 • HSG-MG.A.1 • MODELING WITH GEOMETRY

Model objects with geometric shapes

MAIN	Geometry main question	**HSG-MG.A.1**
BACKWARD 1	Volume formulas	**HSG-GMD.A.3**
BACKWARD 2	Radius from a circumference	**7.G.B.4**
BACKWARD 3	Convert cubic units	**6.RP.A.3.d**
FORWARD 1	Judge a model against a measurement	**HSG-MG.A.1**
FORWARD 2	Use the model to estimate mass with density	**HSG-MG.A.2**

SET 83 • HSG-MG.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A tree trunk is about 6 m tall and has circumference 2.2 m. Model it as a cylinder and estimate its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q1/5

3062

QUESTION ID

S83-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

About 2.3 m³

GRADING NOTE REQUIRED WORK: Find the radius from the circumference, then use $V = \pi r^2 h$.

Rounding to 0.1 required. $r = 2.2/(2\pi) \approx 0.35$ m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pile of sand is shaped like a cone with base diameter 8 m and height 3 m. Estimate its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q2/5

3063

QUESTION ID

S83-M-Q2

TYPE

Short answer

CORRECT ANSWER

50.3 m^3

GRADING NOTE Rounding to 0.1 required. 16π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which shape is the best model for a soup can?

A. Cone

B. Cylinder

C. Square pyramid

D. Sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q3/5

3064

QUESTION ID

S83-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Cylinder

GRADING NOTE Two circular bases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A pool is modeled as a rectangular prism 25 m long, 10 m wide, and 1.8 m deep. How many liters of water does it hold? (1 m^3 = 1,000 L)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q4/5

3065

QUESTION ID
S83-M-Q4

TYPE
Short answer

CORRECT ANSWER
450,000 L

GRADING NOTE Any equivalent form. 450 m^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A pine tree is modeled as a cone. Name one measurement you would take and one way the model differs from the real tree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q5/5

3066

QUESTION ID
S83-M-Q5

TYPE
Short answer

CORRECT ANSWER
For example: height and the width at the bottom of the branches; the real tree has gaps, a trunk, and an uneven outline.

GRADING NOTE Any reasonable measurement and limitation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 1

Volume formulas • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Exact volume of a cylinder with r = 2 and h = 5?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q1/5

3068

QUESTION ID

S83-B1-Q1

TYPE

Short answer

CORRECT ANSWER

20π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact volume of a cone with r = 3 and h = 4?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q2/5

3069

QUESTION ID

S83-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Volume of a 2 × 3 × 4 prism?

A. 52

B. 9

C. 26

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q3/5

3070

QUESTION ID

S83-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 24

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$V = \frac{1}{3}\pi r^2 h$ is the cone formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q4/5

3071

QUESTION ID

S83-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q4/5

True or false?

A sphere's volume is $\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q5/5

3072

QUESTION ID

S83-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4/3\ \pi r^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 2

Radius from a circumference • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A circle has circumference 10π. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q1/5

3074

QUESTION ID

S83-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has circumference 2.2 m. Radius to the nearest hundredth?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q2/5

3075

QUESTION ID
S83-B2-Q2

TYPE
Short answer

CORRECT ANSWER
0.35 m

GRADING NOTE Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

r =

A. C/(2π)

B. πC

C. 2πC

D. C/π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q3/5

3076

QUESTION ID

S83-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. C/(2π)

GRADING NOTE C = 2πr.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q3/5

True or false?

A tape measure around a trunk gives its circumference.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q4/5

3077

QUESTION ID
S83-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Distance around.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q4/5

True or false?

The radius is C/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q5/5

3078

QUESTION ID

S83-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE C/(2π).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 3

Convert cubic units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

ANSWER KEY Reduced copy of the question page, with its answer

How many liters are in 2 m^3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q1/5

3080

QUESTION ID

S83-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2,000 L

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many cubic centimeters are in 1 L?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q2/5

3081

QUESTION ID

S83-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1,000 cm^3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

450 m^3 = ___ L

A. 45,000

B. 0.45

C. 450,000

D. 4,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q3/5

3082

QUESTION ID

S83-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 450,000

GRADING NOTE × 1,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q3/5

3082

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1\ \text{mL} = 1\ \text{cm}^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q4/5

3083

QUESTION ID

S83-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q4/5

3083

True or false?

$1\ m^3 = 100\ L$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q5/5

3084

QUESTION ID

S83-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1,000 L.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q5/5

SET 83 • HSG-MG.A.1

FORWARD 1

Judge a model against a measurement • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A water bottle is modeled as a cylinder with radius 3.8 cm and height 17 cm. The label says 750 mL. Find the model volume to the nearest mL and the percent error.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q1/5

3086

QUESTION ID

S83-F1-Q1

TYPE

Short answer

CORRECT ANSWER

About 771 mL; about 2.8% too high

GRADING NOTE Form required: both (rounding to 1 mL and 0.1%). Percent error = (model − actual)/actual.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A silo is modeled as a cylinder (r = 3 m, h = 12 m) with a hemisphere on top. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q2/5

3087

QUESTION ID

S83-F1-Q2

TYPE

Short answer

CORRECT ANSWER

126π m^3

GRADING NOTE Form required: exact value. $108\pi + 18\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q2/5

True or false?

A model that overestimates by 3% may still be useful for planning.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q3/5

3088

QUESTION ID

S83-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Judgment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A football (pointed ends) is best modeled by

A. two cones joined at their bases

B. a cube

C. a sphere

D. a cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q4/5

3089

QUESTION ID

S83-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. two cones joined at their bases

GRADING NOTE Pointed ends; other answers with reasons may be discussed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q4/5

Why might a cylinder model overestimate the volume of a bottle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q5/5

3090

QUESTION ID
S83-F1-Q5

TYPE
Short answer

CORRECT ANSWER
The real bottle narrows at the neck and has thick walls, so it holds less than the full cylinder.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSG-MG.A.1

FORWARD 2

Use the model to estimate mass with density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A tree trunk modeled as a cylinder has volume about 2.3 m³. The wood has density 600 kg/m³. Estimate the trunk's mass to the nearest 10 kg.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q1/5

3092

QUESTION ID

S83-F2-Q1

TYPE

Short answer

CORRECT ANSWER

About 1390 kg

GRADING NOTE Rounding to the nearest 10 kg required; using 2.3 m³ gives 1,380 kg, also accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q1/5

3092

ANSWER KEY Reduced copy of the question page, with its answer

A sand pile modeled as a cone has volume 16π m³. The sand has density 1,600 kg/m³. Estimate its mass in metric tons (1 t = 1,000 kg), to the nearest ton.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q2/5

3093

QUESTION ID
S83-F2-Q2

TYPE
Short answer

CORRECT ANSWER
80 t

GRADING NOTE Rounding to the nearest ton required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q2/5

3093

True or false?

A more accurate volume model gives a more accurate mass estimate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q3/5

3094

QUESTION ID

S83-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Mass = density × volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q3/5

3094

ANSWER KEY Reduced copy of the question page, with its answer

A pool holds 450 m³ of water (1,000 kg/m³). Mass?

A. 450,000 kg

B. 450 kg

C. 45,000 kg

D. 4,500 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q4/5

3095

GRADING NOTE Product.

QUESTION ID

S83-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 450,000 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how a geometric model and a density work together to estimate the mass of a log.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q5/5

3096

QUESTION ID

S83-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The model gives the volume; mass = density × volume.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q5/5

SET 84 • HSG-MG.A.2 • MODELING WITH GEOMETRY

Apply density based on area

MAIN	Geometry main question	**HSG-MG.A.2**
BACKWARD 1	Unit rates	**6.RP.A.3.b**
BACKWARD 2	Areas of rectangles and circles	**7.G.B.6**
BACKWARD 3	Compare ratios	**7.RP.A.2**
FORWARD 1	Convert area units in density problems	**HSG-MG.A.2**
FORWARD 2	Plan capacity from an area density	**HSG-MG.A.3**

SET 84 • HSG-MG.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A city has 840,000 people and an area of 120 square miles. Find its population density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q1/5

3099

QUESTION ID

S84-M-Q1

TYPE

Short answer

CORRECT ANSWER

7,000 people per square mile

GRADING NOTE Any equivalent form, with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circular forest region has radius 3 km and 4,500 trees. Find the tree density to the nearest tree per square kilometer.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q2/5

3100

QUESTION ID

S84-M-Q2

TYPE

Short answer

CORRECT ANSWER

159 trees per km²

GRADING NOTE Rounding to the nearest whole number required. Area $9\pi \approx 28.27$ km².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q2/5

Town A has 12,000 people on 30 km². Town B has 9,000 people on 20 km². Which town is more densely populated? Show the densities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q3/5

3101

QUESTION ID

S84-M-Q3

TYPE

Short answer

CORRECT ANSWER

Town B (450 per km² vs 400 per km²)

GRADING NOTE Form required: the conclusion with both densities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A region has a population density of 250 people per km² and is a 4 km by 6 km rectangle. Estimate its population.

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q4/5

3102

QUESTION ID
S84-M-Q4

TYPE
Short answer

CORRECT ANSWER
6,000 people

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q4/5

Which is a unit of population density based on area?

A. people per mile

B. people per square mile

C. square miles per person per year

D. people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q5/5

3103

QUESTION ID

S84-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. people per square mile

GRADING NOTE Count ÷ area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 1

Unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

ANSWER KEY Reduced copy of the question page, with its answer

840 items in 12 boxes: items per box?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q1/5

3105

QUESTION ID

S84-B1-Q1

TYPE

Short answer

CORRECT ANSWER

70

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$45 for 9 m² of tile: dollars per square meter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q2/5

3106

QUESTION ID

S84-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$5 per m²

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Unit rate for 300 people on 4 km²?

A. 0.013 km² per person

B. 1,200 people per km²

C. 296 people per km²

D. 75 people per km²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q3/5

3107

QUESTION ID

S84-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 75 people per km²

GRADING NOTE 300 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q3/5

True or false?

A unit rate compares a quantity to one unit of another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q4/5

3108

QUESTION ID

S84-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q4/5

True or false?

600 people on 3 km² is 1,800 people per km².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q5/5

3109

QUESTION ID

S84-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 2

Areas of rectangles and circles • Grade 7 • Geometry • 7.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Area of a 4 km by 6 km rectangle?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q1/5

3111

QUESTION ID

S84-B2-Q1

TYPE

Short answer

CORRECT ANSWER

24 km²

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle with radius 3 km?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q2/5

3112

QUESTION ID

S84-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9π km²

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A square region 5 mi on a side has area

A. 20 mi²

B. 10 mi²

C. 5 mi²

D. 25 mi²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q3/5

3113

QUESTION ID

S84-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 25 mi²

GRADING NOTE 5^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q3/5

3113

True or false?

Area units are square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q4/5

3114

QUESTION ID

S84-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two dimensions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q4/5

True or false?

A circle with radius 3 has area 6π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q5/5

3115

QUESTION ID

S84-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 3

Compare ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Which is greater: 12,000/30 or 9,000/20?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q1/5

3117

QUESTION ID

S84-B3-Q1

TYPE

Short answer

CORRECT ANSWER

9,000/20 (450 > 400)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q1/5

True or false?

To compare densities, compare the unit rates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q2/5

3118

QUESTION ID

S84-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which rate is greatest?

A. 80 per 4 km²

B. 50 per 2 km²

C. 60 per 3 km²

D. 20 per 1 km²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q3/5

3119

QUESTION ID

S84-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 50 per 2 km²

GRADING NOTE 25, 20, 20, 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A has 100 trees on 4 acres; B has 120 trees on 6 acres. Which is denser?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q4/5

3120

QUESTION ID

S84-B3-Q4

TYPE

Short answer

CORRECT ANSWER

A (25 per acre vs 20)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The town with more people is always denser.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q5/5

3121

QUESTION ID

S84-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Depends on area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q5/5

SET 84 • HSG-MG.A.2

FORWARD 1

Convert area units in density problems • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

1 square mile is about 2.59 km². Convert 7,000 people per square mile to people per km² (nearest whole number).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q1/5

3123

QUESTION ID
S84-F1-Q1

TYPE
Short answer

CORRECT ANSWER
2703 people per km²

GRADING NOTE Rounding to the nearest whole number required. Divide by 2.59.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q1/5

3123

A lawn of 400 m² needs grass seed at 35 g per m². How many kilograms of seed are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q2/5

3124

QUESTION ID

S84-F1-Q2

TYPE

Short answer

CORRECT ANSWER

14 kg

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A density in people per km² is a larger number than the same density in people per mi².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q3/5

3125

QUESTION ID

S84-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A km² is smaller, so fewer people per km².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A farm plants 2 trees per 10 m^2. How many trees on 1 hectare (10,000 m^2)?

A. 2,000

B. 200

C. 5,000

D. 20,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q4/5

3126

QUESTION ID

S84-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2,000

GRADING NOTE 0.2 per m^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A park of 3 hectares (1 ha = 10,000 m²) has 5,400 visitors. Visitors per 100 m²?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q5/5

3127

QUESTION ID

S84-F1-Q5

TYPE

Short answer

CORRECT ANSWER

18

GRADING NOTE Any equivalent form. 300 units of 100 m².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q5/5

SET 84 • HSG-MG.A.2

FORWARD 2

Plan capacity from an area density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Safety rules allow 2 people per m^2 in a plaza. The plaza is a 30 m by 40 m rectangle. Maximum crowd?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q1/5

3129

QUESTION ID
S84-F2-Q1

TYPE
Short answer

CORRECT ANSWER
2,400 people

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A festival expects 5,000 people and wants at most 1.5 people per m². What is the smallest area needed (nearest m²)?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q2/5

3130

QUESTION ID

S84-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3334 m²

GRADING NOTE Rounding up required: 3,334 m² (3,333.3… rounded up so the limit is met).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q2/5

True or false?

When planning for a safety limit, round the capacity up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q3/5

3131

QUESTION ID

S84-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Round capacity down (and area up) to stay within the limit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q3/5

A circular dance floor of radius 5 m allows 3 people per m². Capacity is about

A. 235 people

B. 471 people

C. 75 people

D. 47 people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q4/5

3132

QUESTION ID

S84-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 235 people

GRADING NOTE $75\pi \approx 235.6$, round down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A circular dance floor of radius 5 m allows at most 3 people per m², and $3 \cdot 25\pi \approx 235.6$. Explain why the capacity is 235 people, not 236.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q5/5

3133

QUESTION ID
S84-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Rounding up would exceed the allowed density.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q5/5

SET 85 • HSG-MG.A.2 • MODELING WITH GEOMETRY

Apply density based on volume

MAIN	Geometry main question	**HSG-MG.A.2**
BACKWARD 1	Volume of prisms and cylinders	**HSG-GMD.A.3**
BACKWARD 2	Unit rates	**6.RP.A.3.b**
BACKWARD 3	Products of numbers in scientific notation	**8.EE.A.4**
FORWARD 1	Use density to identify materials and find volumes	**HSG-MG.A.2**
FORWARD 2	Design within a mass limit	**HSG-MG.A.3**

SET 85 • HSG-MG.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

An aluminum block measures 2 cm by 3 cm by 5 cm and has mass 81 g. Find its density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q1/5

3136

QUESTION ID

S85-M-Q1

TYPE

Short answer

CORRECT ANSWER

2.7 g/cm³

GRADING NOTE Any equivalent form, with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylindrical tank (radius 0.5 m, height 2 m) is full of water (1,000 kg/m³). Find the mass of the water to the nearest kilogram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q2/5

3137

QUESTION ID

S85-M-Q2

TYPE

Short answer

CORRECT ANSWER

1571 kg

GRADING NOTE Rounding to the nearest kilogram required. 0.5π m³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q2/5

A heater must supply 20 BTU per cubic foot of room. A room is 12 ft by 15 ft with an 8 ft ceiling. How many BTU are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q3/5

3138

QUESTION ID

S85-M-Q3

TYPE

Short answer

CORRECT ANSWER

28,800 BTU

GRADING NOTE Any equivalent form. 1,440 ft^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A sample has 3×10^6 bacteria per mL. How many bacteria are in 250 mL?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q4/5

3139

QUESTION ID
S85-M-Q4

TYPE
Short answer

CORRECT ANSWER
7.5×10^8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Water has density 1 g/cm³. Which object floats in water?

A. A 1,000 g brick with volume 400 cm³

B. A 600 g block with volume 500 cm³

C. A 100 g ball with volume 50 cm³

D. A 400 g block with volume 500 cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q5/5

3140

QUESTION ID

S85-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. A 400 g block with volume 500 cm³

GRADING NOTE Density less than 1 g/cm³ (0.8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 1

Volume of prisms and cylinders • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Volume of a 2 × 3 × 5 cm block?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q1/5

3142

QUESTION ID

S85-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30 cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact volume of a cylinder with r = 0.5 m, h = 2 m?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q2/5

3143

QUESTION ID

S85-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.5π m^3

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Volume of a 12 × 15 × 8 ft room?

A. 1,440 ft³

B. 180 ft³

C. 2,880 ft³

D. 35 ft³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q3/5

3144

QUESTION ID

S85-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1,440 ft³

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q3/5

True or false?

Volume is needed before a volume density can be used.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q4/5

3145

QUESTION ID
S85-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Density = amount ÷ volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A 1 m cube has volume 3 m³.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q5/5

3146

QUESTION ID

S85-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1 m³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 2

Unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

ANSWER KEY Reduced copy of the question page, with its answer

81 g in 30 cm^3: grams per cm^3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q1/5

3148

QUESTION ID

S85-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2.7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

20 BTU per ft^3 for 100 ft^3: total?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q2/5

3149

QUESTION ID

S85-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2,000 BTU

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

500 g in 250 cm³ is

A. 250 g/cm³

B. 2 g/cm³

C. 0.5 g/cm³

D. 750 g/cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q3/5

3150

QUESTION ID

S85-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2 g/cm³

GRADING NOTE Divide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q3/5

True or false?

Amount = rate × quantity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q4/5

3151

QUESTION ID

S85-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rate use.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q4/5

True or false?

Density = volume ÷ mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q5/5

3152

QUESTION ID

S85-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Mass ÷ volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 3

Products of numbers in scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$(3 \times 10^6) \times 250$ in scientific notation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q1/5

3154

QUESTION ID

S85-B3-Q1

TYPE

Short answer

CORRECT ANSWER

7.5×10^8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 2,400,000 in scientific notation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q2/5

3155

QUESTION ID

S85-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2.4×10^6

GRADING NOTE Form required: scientific notation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(2 \times 10^3)(4 \times 10^2) =$

A. 6×10^5

B. 8×10^5

C. 8×10^6

D. 8×10^1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q3/5

3156

QUESTION ID

S85-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8×10^5

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q3/5

True or false?

$10^3 \times 10^2 = 10^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q4/5

3157

QUESTION ID

S85-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Exponent rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$5 \times 10^2 \times 3 = 15 \times 10^6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q5/5

3158

QUESTION ID

S85-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1.5×10^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q5/5

SET 85 • HSG-MG.A.2

FORWARD 1

Use density to identify materials and find volumes • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A ring has mass 9.65 g and volume 0.5 cm³. Gold has density 19.3 g/cm³. Could the ring be pure gold?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q1/5

3160

QUESTION ID

S85-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes:

9.65 ÷ 0.5 = 19.3 g/cm³

GRADING NOTE Form required: the conclusion with the density.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 1,350 g piece of aluminum (2.7 g/cm³) is melted into a cube. Find the edge of the cube, to the nearest tenth of a centimeter.

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q2/5

3161

QUESTION ID

S85-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7.9 cm

GRADING NOTE Rounding to 0.1 required. Volume 500 cm³; edge $\sqrt[3]{500}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q2/5

True or false?

Two objects with equal mass but different volumes have different densities.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q3/5

3162

QUESTION ID

S85-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Density = m/V.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Oil (0.9 g/cm³) in a 2,000 cm³ jug has mass

A. 1,800 g

B. 2,000 g

C. 2,222 g

D. 1,100 g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q4/5

3163

QUESTION ID

S85-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1,800 g

GRADING NOTE 0.9 × 2,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q4/5

A balloon holds 0.02 m^3 of helium (0.18 kg/m^3). Mass of the helium in grams?

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q5/5

3164

QUESTION ID

S85-F1-Q5

TYPE

Short answer

CORRECT ANSWER

3.6 g

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q5/5

SET 85 • HSG-MG.A.2

FORWARD 2

Design within a mass limit • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A truck can carry 12,000 kg. Gravel has density 1,500 kg/m³. What is the largest volume of gravel it can carry?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q1/5

3166

QUESTION ID

S85-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8 m³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A shelf holds at most 40 kg. Books have density 0.8 g/cm³. How many cubic centimeters of books can it hold?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q2/5

3167

QUESTION ID
S85-F2-Q2

TYPE
Short answer

CORRECT ANSWER
50,000 cm³

GRADING NOTE Any equivalent form. 40,000 g ÷ 0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q2/5

True or false?

Mass limits become volume limits by multiplying by the density.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q3/5

3168

QUESTION ID

S85-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide by the density: V = m/d.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q3/5

A cylindrical water tank must weigh at most 3,140 kg when full (water 1,000 kg/m³, ignore the tank). With radius 1 m, its height can be at most about

A. 10 m

B. 3.14 m

C. 1 m

D. 0.32 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q4/5

3169

QUESTION ID

S85-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 m

GRADING NOTE 3.14 m³ ÷ π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A concrete block must be under 25 kg. Concrete is 2,400 kg/m³. Will a 20 cm × 20 cm × 30 cm block work?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q5/5

3170

QUESTION ID

S85-F2-Q5

TYPE

Short answer

CORRECT ANSWER

No: 0.012 m³ × 2,400 kg/m³ = 28.8 kg, which exceeds 25 kg.

GRADING NOTE Form required: the conclusion with the mass.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q5/5

SET 86 • HSG-MG.A.3 • MODELING WITH GEOMETRY

Design to satisfy physical constraints

MAIN	Geometry main question	**HSG-MG.A.3**
BACKWARD 1	Volume of a rectangular prism	**7.G.B.6**
BACKWARD 2	Use the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Ratios and rates	**7.RP.A.2**
BACKWARD 4	Inequalities as limits	**7.EE.B.4**
FORWARD 1	Satisfy several constraints at once	**HSG-MG.A.3**
FORWARD 2	Find the best design with a function graph	**HSF-IF.C.7.c**

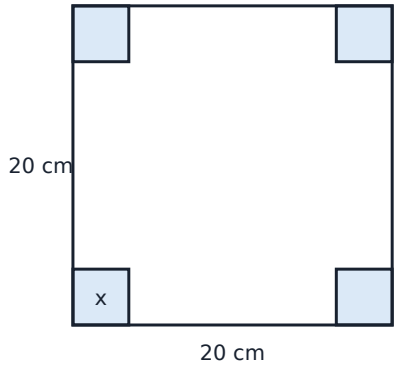
SET 86 • HSG-MG.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Compare the volumes for x = 2, 3, and 4. Which gives the largest box?



Not drawn to scale.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q1/5

3173

QUESTION ID
S86-M-Q1

TYPE
Short answer

CORRECT ANSWER
x = 2: 512 cm³; x = 3: 588 cm³; x = 4: 576 cm³; x = 3 is largest

GRADING NOTE Form required: the three volumes and the choice. $V = x(20 - 2x)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Accessibility rules allow a ramp slope of at most 1 : 12 (1 unit of rise per 12 units of run). A ramp must rise 30 in. What is the shortest horizontal run allowed, in feet?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q2/5

3174

QUESTION ID
S86-M-Q2

TYPE
Short answer

CORRECT ANSWER
30 ft

GRADING NOTE Any equivalent form. 30 × 12 = 360 in.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q2/5

A rectangular pen will use 40 m of fencing. Compare a 10 m by 10 m pen with a 5 m by 15 m pen. Which encloses more area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q3/5

3175

QUESTION ID

S86-M-Q3

TYPE

Short answer

CORRECT ANSWER

The 10 m by 10 m pen (100 m² vs 75 m²)

GRADING NOTE Form required: the choice with both areas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylindrical tank must fit in a closet 2 m wide, 2 m deep, and 3 m tall. What is the largest possible volume, to the nearest tenth of a cubic meter?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q4/5

3176

QUESTION ID

S86-M-Q4

TYPE

Short answer

CORRECT ANSWER

9.4 m³

GRADING NOTE Rounding to 0.1 required. Radius 1 m, height 3 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A 13 ft ladder leans on a wall. For safety its base must be at least 5 ft from the wall. The highest point it can reach is

A. 18 ft

B. 12 ft

C. 13 ft

D. 8 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q5/5

3177

QUESTION ID

S86-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 12 ft

GRADING NOTE √(169 − 25).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 1

Volume of a rectangular prism • Grade 7 • Geometry • 7.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Volume of a box 14 cm by 14 cm by 3 cm?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q1/5

3179

QUESTION ID

S86-B1-Q1

TYPE

Short answer

CORRECT ANSWER

588 cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 20 cm side with 3 cm cut from each end leaves

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q2/5

3180

QUESTION ID

S86-B1-Q2

TYPE

Short answer

CORRECT ANSWER

14 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q2/5

Volume of a box 12 by 12 by 4?

A. 28

B. 576

C. 48

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q3/5

3181

QUESTION ID

S86-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 576

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q3/5

True or false?

Cutting larger corners makes the box taller but its base smaller.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q4/5

3182

QUESTION ID

S86-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Trade-off.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q4/5

True or false?

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner. The box's base side is 20 − x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q5/5

3183

QUESTION ID
S86-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 20 − 2x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 2

Use the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse 13, leg 5: other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q1/5

3185

QUESTION ID

S86-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Legs 30 in. and 360 in.: hypotenuse to the nearest inch?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q2/5

3186

QUESTION ID

S86-B2-Q2

TYPE

Short answer

CORRECT ANSWER

361 in.

GRADING NOTE Rounding to the nearest inch required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q2/5

A ladder, wall, and ground form a

A. right triangle

B. circle

C. square

D. equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q3/5

3187

QUESTION ID

S86-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. right triangle

GRADING NOTE Wall ⊥ ground.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q3/5

True or false?

A ladder leans against a vertical wall on level ground. The ladder is the hypotenuse of the right triangle formed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q4/5

3188

QUESTION ID

S86-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Opposite the right angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q4/5

True or false?

The ladder's height on the wall is 13 + 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q5/5

3189

QUESTION ID

S86-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use the theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 3

Ratios and rates • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A ratio of 1 : 12 for a rise of 2 in. gives what run?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q1/5

3191

QUESTION ID

S86-B3-Q1

TYPE

Short answer

CORRECT ANSWER

24 in.

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is a slope of 3 : 30 steeper than 1 : 12?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q2/5

3192

QUESTION ID

S86-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: 3 : 30 = 1 : 10, and 1/10 > 1/12.

GRADING NOTE Form required: yes/no with the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q2/5

3192

ANSWER KEY Reduced copy of the question page, with its answer

360 in. =

A. 30 ft

B. 36 ft

C. 12 ft

D. 3.6 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q3/5

3193

GRADING NOTE ÷ 12.

QUESTION ID

S86-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 30 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A ratio of 1 : 12 means 1 inch up for every 12 inches across.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q4/5

3194

QUESTION ID

S86-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q4/5

True or false?

A 1 : 8 ramp is less steep than a 1 : 12 ramp.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q5/5

3195

QUESTION ID

S86-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Steeper.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 4

Inequalities as limits • Grade 7 • Expressions & Equations • 7.EE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The run r must satisfy r ≥ 360. Is r = 300 allowed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q1/5

3197

QUESTION ID

S86-B4-Q1

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A diameter d must satisfy d ≤ 2. Largest radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q2/5

3198

QUESTION ID

S86-B4-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

"At most 3 m" means

A. ≥ 3

B. < 3 only

C. ≤ 3

D. = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q3/5

3199

QUESTION ID

S86-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. ≤ 3

GRADING NOTE Inequality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

"At least 5 ft" allows exactly 5 ft.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q4/5

3200

QUESTION ID

S86-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ≥.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q4/5